



UNIVERSITY OF PISA
MASTER'S DEGREE IN COMPUTER SCIENCE,
ARTIFICIAL INTELLIGENCE

**Computational Mathematics for Learning &
Data Analysis (646AA)**

Group number: 63
Project number: 25 ML

Members:
Matthew Bernardi
Gabriele Marsili

ACADEMIC YEAR: 2025/2026

Project statement.

25. (P) is so-called *extreme learning*, i.e., a neural network with one hidden layer, $y = w\sigma(W_1x)$, where the weight matrix for the hidden layer W_1 is a fixed random matrix, $\sigma(\cdot)$ is an elementwise activation function of your choice, and the output weight vector w is chosen by solving a least-squares problem with L_1 regularization $\arg \min_w f(w)$, with $f(w) = \frac{1}{2}\|Xw - y\|_2^2 + \lambda\|w\|_1$.

(A1) is *iteratively reweighted least squares*: i.e., an iteration where you solve at each step the least-squares problem $w_{k+1} = \arg \min_w f_k(w)$, $f_k(w) = \frac{1}{2}\|Xw - y\|_2^2 + \lambda\|W_k w\|_2^2$ where W_k is the diagonal matrix with entries $(W_k)_{ii} = |(w_k)_i|^{-1/2}$.

(A2) is an algorithm of the class of the *deflected subgradient methods*.

Contents

1	Introduction to the problem	2
1.1	Extreme Learning Machine	2
1.2	Least-Squares Problem	2
1.3	Regularization Techniques	4
1.3.1	L_1 regularization	4
1.4	Objective Function Analysis	4
1.4.1	Convexity	4
1.4.2	Non-smoothness	5
1.4.3	Optimality conditions	5
2	Algorithm 1: Iteratively Reweighted Least Squares	6
2.1	The core idea: a strict upper bound for the L_1 norm	6
2.2	The Majorization-Minimization interpretation	7
2.3	The weighted least-squares subproblem	8
2.4	The complete algorithm	9
2.5	Convergence analysis	10
2.6	Computational cost	11
2.7	Expected behavior on our problem	12
3	Algorithm 2: Deflected Subgradient Method	13
3.1	Why the plain subgradient method is inadequate	13
3.1.1	The deflection mechanism	13
3.2	Convergence framework: balancing deflection and stepsize	14
3.2.1	Optimal deflection parameter	14
3.3	The target-level mechanism	15
3.4	The complete algorithm	15
3.4.1	Parameter calibration for ELM LASSO	15
3.5	Application to the ELM LASSO problem	18
3.5.1	Subgradient computation for ELM LASSO	18
3.5.2	Convergence analysis and hypothesis verification for ELM LASSO	18
3.6	Expected practical behavior	21

4	Algorithm Comparison	22
4.1	Core strategy: how each algorithm handles non-smoothness . . .	22
4.2	Convergence behaviour	23
4.2.1	Monotonicity	23
4.2.2	Rate	23
4.2.3	Convergence on the ELM problem	23
4.3	Computational cost	23
4.3.1	Per-iteration cost	23
4.3.2	Total cost	24
4.4	Solution quality	24
4.4.1	Sparsity	24
4.4.2	Accuracy	25
4.5	Comparison summary for the ELM problem	25
5	Experimental Results	26
5.1	Experimental setup	26
5.1.1	Datasets	26
5.1.2	Implementation note: from the first submission to the theory-pure rule	28
5.2	Convergence behaviour	30
5.3	Parameter sensitivity	32
5.3.1	IRLS: ε_{thr} and λ_{LASSO}	32
5.3.2	IRLS: linear solver choice (Cholesky vs. Conjugate Gra- dient)	33
5.3.3	SGPTL: δ_0 and ρ	34
5.4	Solution quality and sparsity	38
5.5	Scalability	39
5.6	Iterations to a target accuracy	40
5.7	Validation on real datasets	41
6	Conclusions	45

Chapter 1

Introduction to the problem

1.1 Extreme Learning Machine

Extreme Learning Machine (ELM) [HUANG2006489] is a machine learning model implemented as a neural network with a single hidden layer, in which:

- $\mathbf{W}_1 \in \mathbb{R}^{H \times N}$ is the input-to-hidden weight matrix (where H is the number of nodes in the hidden layer and N is the number of input nodes) that is **randomly generated** and **static** (does not change during training) [HUANG2006489].
- $\sigma(\cdot)$ is the element-wise (nonlinear) activation function applied to the $\mathbf{W}_1 \mathbf{x} \in \mathbb{R}^H$ vector, where $\mathbf{x} \in \mathbb{R}^N$ is the input vector. The choice of using an activation function for the hidden layer results, under mild assumptions and with high probability, in the hidden layer feature matrix with full rank when $M \geq H$, since random projections through a nonlinear function produce linearly independent features.
- $\mathbf{w} \in \mathbb{R}^H$ is the hidden-to-output weight vector.
- $\hat{y} = \mathbf{w}^T \sigma(\mathbf{W}_1 \mathbf{x}) \in \mathbb{R}$ is the prediction of the model.

Once $\mathbf{W}_1$ is fixed, knowing that $\mathbf{X}_{\text{origin}} \in \mathbb{R}^{M \times N}$ is the matrix of all the inputs, the hidden-layer activations $\mathbf{X} = \sigma(\mathbf{X}_{\text{origin}} \mathbf{W}_1^T)$ are constant, and finding $\mathbf{w}$ reduces to solving the Least-Squares problem described in the following section.

1.2 Least-Squares Problem

The definition of the Least Squares Problem [cortinovis-introleastsquares] in the ELM case is the following:

$$\min_{\mathbf{w}} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 \quad (1.1)$$

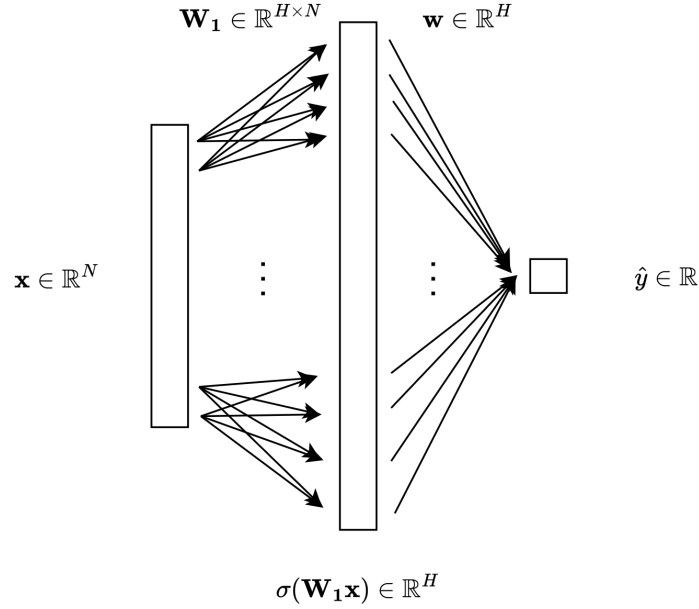


Figure 1.1: Visual representation of an Extreme Learning Machine model.

The Least-Squares problem always admits at least one solution, and the solution is unique if and only if the matrix $\mathbf{X}$ has full column rank. In that case, we can write the optimization problem as follows:

$$\min_{\mathbf{w}} \frac{1}{2} \mathbf{w}^T \mathbf{X}^T \mathbf{X} \mathbf{w} - \mathbf{y}^T \mathbf{X} \mathbf{w} + \frac{1}{2} \mathbf{y}^T \mathbf{y} \quad (1.2)$$

since the gradient of this function is $\mathbf{X}^T \mathbf{X} \mathbf{w} - \mathbf{X}^T \mathbf{y}$, the Hessian matrix is $\mathbf{X}^T \mathbf{X} \succ 0$, which means that the function is strictly convex. In this case, the minimum exists and is unique, and can be found solving the equation:

$$\mathbf{X}^T \mathbf{X} \mathbf{w} - \mathbf{X}^T \mathbf{y} = 0$$

where $\mathbf{X}^T \mathbf{X}$ is square invertible since it's positive definite. The unique solution is $\mathbf{w}_0 = (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$. However, adding the L_1 regularization term (Section 1.3) destroys the closed-form structure, requiring iterative methods.

1.3 Regularization Techniques

1.3.1 L_1 regularization

When adding the L_1 regularization term to the Least-Squares, the equation is the following:

$$\min_{\mathbf{w}} f(\mathbf{w}) = \frac{1}{2} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \lambda_{\text{LASSO}} \sum_i^H |w_i| \quad (1.3)$$

This form is particularly nice since it pushes the weights $w_i \rightarrow 0$, but the price we have to pay is the non-differentiability of the function where $w_i = 0$, which prevents direct use of gradient-based methods.

Because of that, we can use the Deflected Subgradient Methods (2) and the IRLS algorithm (2), which we will see in the following chapters.

1.4 Objective Function Analysis

Let's break down the function to minimize in two parts:

$$\min_{\mathbf{w}} f(\mathbf{w}) = \underbrace{\frac{1}{2} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2}_{g(\mathbf{w})} + \underbrace{\lambda_{\text{LASSO}} \|\mathbf{w}\|_1}_{h(\mathbf{w})}$$

It is useful to analyse the mathematical properties of $f(\mathbf{w})$, as they determine which algorithms can be applied.

1.4.1 Convexity

The mathematical properties of the objective function $f(\mathbf{w})$ are heavily dependent on the matrix $\mathbf{X}$. As we establish below, a full-rank matrix provides the strictly convex structure necessary to guarantee a unique solution.

Proposition 1.1. *If $\mathbf{X} \in \mathbb{R}^{M \times H}$ has full column rank, then $f(\mathbf{w}) = \frac{1}{2} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \lambda_{\text{LASSO}} \|\mathbf{w}\|_1$ is strictly convex and admits a unique global minimizer $\mathbf{w}^*$ [Boyd-Vandenberghe'2004].*

Proof. Strict convexity. Since $\mathbf{X}$ has full column rank, for any $\mathbf{v} \neq \mathbf{0}$ we have $\mathbf{X}\mathbf{v} \neq \mathbf{0}$, so $\mathbf{v}^T (\mathbf{X}^T \mathbf{X}) \mathbf{v} = \|\mathbf{X}\mathbf{v}\|_2^2 > 0$. Hence the Hessian of g is $\nabla^2 g = \mathbf{X}^T \mathbf{X} \succ 0$, making g strictly convex. h is convex as a non-negative multiple of a norm. The sum of a strictly convex function and a convex function is strictly convex [frangioni-slides-unconstrained], so f is strictly convex.

Existence and uniqueness of the minimizer. Since $\mathbf{X}^T \mathbf{X} \succ 0$, g is coercive: $g(\mathbf{w}) \rightarrow +\infty$ as $\|\mathbf{w}\| \rightarrow \infty$. Because $h \geq 0$, $f = g + h$ is also coercive. This means the sub-level set $\{\mathbf{w} : f(\mathbf{w}) \leq f(\mathbf{0})\}$ is compact, and since f is continuous, it attains its minimum on this set. Strict convexity then rules out two distinct minimizers: if $\mathbf{w}_1 \neq \mathbf{w}_2$ both minimized f , the midpoint $\frac{\mathbf{w}_1 + \mathbf{w}_2}{2}$ would give a strictly smaller value, a contradiction. $\square$

1.4.2 Non-smoothness

Unlike the quadratic term, the L_1 penalty is **not differentiable** [frangioni-slides-nonsmooth] at any point where some component $w_i = 0$, so f is non-smooth. At points where all $w_i \neq 0$, the gradient exists and is:

$$\nabla f(\mathbf{w}) = \mathbf{X}^T(\mathbf{X}\mathbf{w} - \mathbf{y}) + \lambda_{\text{LASSO}} \text{sign}(\mathbf{w})$$

where $\text{sign}(\mathbf{w})$ is applied element-wise. At points where some $w_i = 0$, one must reason in terms of *subgradients* (see Chapter 2).

1.4.3 Optimality conditions

The optimality condition is expressed via the **subdifferential** [frangioni-slides-nonsmooth]: $\mathbf{w}^*$ is a global minimizer for the problem if and only if:

$$0 \in \partial f(\mathbf{w}^*) = \mathbf{X}^T(\mathbf{X}\mathbf{w}^* - \mathbf{y}) + \lambda_{\text{LASSO}} \partial \|\mathbf{w}^*\|_1$$

where the subdifferential of the L_1 norm is given component-wise by:

$$[\partial \|\mathbf{w}\|_1]_i = \begin{cases} \{+1\} & \text{if } w_i > 0, \\ \{-1\} & \text{if } w_i < 0, \\ [-1, +1] & \text{if } w_i = 0. \end{cases}$$

This condition can be written component-wise as: for each $i = 1, \dots, H$,

$$[\mathbf{X}^T(\mathbf{X}\mathbf{w}^* - \mathbf{y})]_i + \lambda_{\text{LASSO}} s_i = 0, \quad s_i \in \partial |w_i^*|. \quad (1.4)$$

Chapter 2

Algorithm 1: Iteratively Reweighted Least Squares

Iteratively Reweighted Least Squares (IRLS) [daubechies2008iterativelyreweightedleast-squaresminimization] is a classical algorithm for solving optimization problems involving non-smooth objective functions of the form of a p -norm. Its core idea is to handle non-smoothness by solving a sequence of *weighted* least-squares problems, each of which is smooth and admits an efficient closed-form solution. In our setting, IRLS is used to minimize the L_1 -regularized least-squares objective:

$$\min_{\mathbf{w} \in \mathbb{R}^H} f(\mathbf{w}) = \underbrace{\frac{1}{2} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2}_{g(\mathbf{w})} + \underbrace{\lambda_{\text{LASSO}} \|\mathbf{w}\|_1}_{h(\mathbf{w})} \quad (2.1)$$

where $\mathbf{X} \in \mathbb{R}^{M \times H}$, $\mathbf{y} \in \mathbb{R}^M$, and $\lambda_{\text{LASSO}} > 0$ is the regularization hyperparameter of the original problem.

2.1 The core idea: a strict upper bound for the L_1 norm

The difficulty in (2.1) is the L_1 term: convex but non-differentiable at every component crossing zero. To replace it with a smooth surrogate we use the elementary inequality $(|w_i| - |(w_k)_i|)^2 \geq 0$, which after expansion and division by $2|(w_k)_i|$ gives the majorization

$$|w_i| \leq \frac{w_i^2}{2|(w_k)_i|} + \frac{|(w_k)_i|}{2}, \quad (w_k)_i \neq 0. \quad (2.2)$$

Summing over $i = 1, \dots, H$ yields the bound on the L_1 norm

$$\|\mathbf{w}\|_1 \leq \frac{1}{2} \mathbf{w}^T \mathbf{W}_k^T \mathbf{W}_k \mathbf{w} + \frac{1}{2} \|\mathbf{w}_k\|_1 \quad (2.3)$$

where $\mathbf{W}_k \in \mathbb{R}^{H \times H}$ is a diagonal matrix with entries:

$$(\mathbf{W}_k)_{ii} = |(w_k)_i|^{-1/2} \quad (2.4)$$

This bounds the non-smooth L_1 term using a *smooth, quadratic* expression $\frac{1}{2} \mathbf{w}^T \mathbf{W}_k^T \mathbf{W}_k \mathbf{w}$, plus a constant term that depends only on the current iterate $\mathbf{w}_k$. Intuitively, $\mathbf{W}_k^T \mathbf{W}_k$ implements a *personalized penalization* of each component of $\mathbf{w}$ [chartrand2008iteratively]: components with small magnitude at iteration k receive a large weight $|(w_k)_i|^{-1}$ and are strongly pushed toward zero, while components with large magnitude receive a small weight and are left relatively free. This *reweighting* mechanism is what drives IRLS to produce sparse iterates, mimicking the sparsity-inducing effect of the L_1 norm through a sequence of smooth quadratic problems.

Handling near-zero components. Equation (2.4) becomes numerically unstable when $(w_k)_i \approx 0$, since $|(w_k)_i|^{-1/2}$ can blow up. To prevent division by (near) zero, a **safety threshold** $\varepsilon_{\text{thr}} > 0$ is introduced:

$$(\mathbf{W}_k)_{ii} = \left(\max(|(w_k)_i|, \varepsilon_{\text{thr}}) \right)^{-1/2} \quad (2.5)$$

Typical values are $\varepsilon_{\text{thr}} \in [10^{-6}, 10^{-10}]$.

2.2 The Majorization-Minimization interpretation

IRLS fits naturally inside the Majorization-Minimization (MM) framework [nguyen2016introductionmmalg]. Instead of minimizing f directly, at each iteration k we build a *surrogate* $Q(\mathbf{w}, \mathbf{w}_k)$ that (i) majorizes f , i.e. $Q(\mathbf{w}, \mathbf{w}_k) \geq f(\mathbf{w})$ for all $\mathbf{w}$; (ii) touches f at $\mathbf{w}_k$, i.e. $Q(\mathbf{w}_k, \mathbf{w}_k) = f(\mathbf{w}_k)$; and (iii) is smooth, so that it can be minimized in closed form. Plugging the quadratic upper bound (2.3) into (2.1) gives our surrogate,

$$Q(\mathbf{w}, \mathbf{w}_k) = \frac{1}{2} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \frac{\lambda_{\text{LASSO}}}{2} \|\mathbf{W}_k \mathbf{w}\|_2^2 + \frac{\lambda_{\text{LASSO}}}{2} \|\mathbf{w}_k\|_1, \quad (2.6)$$

which is indeed a quadratic in $\mathbf{w}$. The majorization property is built in via (2.2). The touching property follows from a short computation: at $\mathbf{w} = \mathbf{w}_k$, each summand of the quadratic penalty becomes $(\mathbf{W}_k)_{ii}^2 (w_k)_i^2 = (w_k)_i^2 / |(w_k)_i| = |(w_k)_i|$, so

$$Q(\mathbf{w}_k, \mathbf{w}_k) = \frac{1}{2} \|\mathbf{X}\mathbf{w}_k - \mathbf{y}\|_2^2 + \frac{\lambda_{\text{LASSO}}}{2} \|\mathbf{w}_k\|_1 + \frac{\lambda_{\text{LASSO}}}{2} \|\mathbf{w}_k\|_1 = f(\mathbf{w}_k).$$

Taking $\mathbf{w}_{k+1} = \arg \min_{\mathbf{w}} Q(\mathbf{w}, \mathbf{w}_k)$, the two MM properties combined give $f(\mathbf{w}_{k+1}) \leq Q(\mathbf{w}_{k+1}, \mathbf{w}_k) \leq Q(\mathbf{w}_k, \mathbf{w}_k) = f(\mathbf{w}_k)$, so the sequence $\{f(\mathbf{w}_k)\}$ is non-increasing. IRLS thus guarantees a monotone decrease of the objective in exact arithmetic; in floating-point arithmetic the monotonicity is preserved up to the accuracy of the linear solver used for the subproblem.

2.3 The weighted least-squares subproblem

To find the next iterate $\mathbf{w}_{k+1}$, we must minimize the surrogate function $Q(\mathbf{w}, \mathbf{w}_k)$ with respect to $\mathbf{w}$:

$$\mathbf{w}_{k+1} = \arg \min_{\mathbf{w}} \left(\frac{1}{2} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \frac{\lambda_{\text{LASSO}}}{2} \|\mathbf{W}_k \mathbf{w}\|_2^2 + \frac{\lambda_{\text{LASSO}}}{2} \|\mathbf{w}_k\|_1 \right)$$

Since the term $\frac{\lambda_{\text{LASSO}}}{2} \|\mathbf{w}_k\|_1$ is constant with respect to $\mathbf{w}$, it does not affect the optimization and can be dropped entirely. The minimization problem simplifies to a standard **weighted least-squares problem with L_2 regularization** (Ridge regression):

$$\mathbf{w}_{k+1} = \arg \min_{\mathbf{w}} \left(\frac{1}{2} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \lambda_{\text{IRLS}} \|\mathbf{W}_k \mathbf{w}\|_2^2 \right) \quad (2.7)$$

provided that we strictly define the IRLS regularization parameter as:

$$\lambda_{\text{IRLS}} = \frac{1}{2} \lambda_{\text{LASSO}} \quad (2.8)$$

Relation to the LASSO regularization parameter.

The factor $1/2$ in (2.8) is forced by the majorization constant in (2.3). The upper bound on $\|\mathbf{w}\|_1$ carries a coefficient $\frac{1}{2}$ in front of the quadratic form $\mathbf{w}^T \mathbf{W}_k^T \mathbf{W}_k \mathbf{w}$, so the penalty entering the surrogate (2.6) is $\frac{\lambda_{\text{LASSO}}}{2} \|\mathbf{W}_k \mathbf{w}\|_2^2$. Rewriting this as $\lambda_{\text{IRLS}} \|\mathbf{W}_k \mathbf{w}\|_2^2$ in (2.7) therefore requires $\lambda_{\text{IRLS}} = \frac{1}{2} \lambda_{\text{LASSO}}$. This is precisely the factor that makes the fixed point of IRLS coincide with the LASSO optimum, as shown in the proof of Theorem 2.2.

Proposition 2.1 (Weighted normal equations). *The unique solution of (2.7) satisfies:*

$$(\mathbf{X}^T \mathbf{X} + 2\lambda_{\text{IRLS}} \mathbf{W}_k^T \mathbf{W}_k) \mathbf{w}_{k+1} = \mathbf{X}^T \mathbf{y} \quad (2.9)$$

Proof. The gradient of the objective in (2.7) is:

$$\nabla f_k(\mathbf{w}) = \mathbf{X}^T (\mathbf{X}\mathbf{w} - \mathbf{y}) + 2\lambda_{\text{IRLS}} \mathbf{W}_k^T \mathbf{W}_k \mathbf{w}$$

Setting this to zero yields (2.9). The coefficient matrix $\mathbf{X}^T \mathbf{X} + 2\lambda_{\text{IRLS}} \mathbf{W}_k^T \mathbf{W}_k$ is symmetric positive definite (it is the sum of $\mathbf{X}^T \mathbf{X} \succeq 0$ and $2\lambda_{\text{IRLS}} \mathbf{W}_k^T \mathbf{W}_k \succ 0$ since $\varepsilon_{\text{thr}} > 0$), so the system has a unique solution. $\square$

Since $\mathbf{W}_k$ is diagonal, defining $\mathbf{Q}_k = \mathbf{X}^T \mathbf{X} + 2\lambda_{\text{IRLS}} \mathbf{W}_k^T \mathbf{W}_k$ and $\mathbf{b} = \mathbf{X}^T \mathbf{y}$, equation (2.9) reduces to:

$$\mathbf{Q}_k \mathbf{w}_{k+1} = \mathbf{b} \quad (2.10)$$

This is an $H \times H$ symmetric positive definite (SPD) linear system, which can be solved efficiently at each iteration. Note that $\mathbf{b} = \mathbf{X}^T \mathbf{y}$ can be pre-computed once before the iterative loop, since it does not depend on k . Similarly, $\mathbf{A} = \mathbf{X}^T \mathbf{X}$ is pre-computed once, and only the diagonal $2\lambda_{\text{IRLS}} \mathbf{W}_k^T \mathbf{W}_k$ is updated at each iteration.

Solving the linear system. Because $\mathbf{Q}_k$ is SPD, two efficient methods can be used:

- **Cholesky factorization:** [trefethen1997numerical] computes $\mathbf{Q}_k = \mathbf{L}\mathbf{L}^T$ in $O(H^3/3)$ operations, then solves two triangular systems in $O(H^2)$. This is the direct method of choice for moderate-sized problems.
- **Conjugate Gradient (CG):** [trefethen1997numerical] an iterative method that converges in at most H steps in exact arithmetic, with convergence rate depending on the condition number $\kappa(\mathbf{Q}_k)$. Preferable for large-scale problems where H is too large for Cholesky.

2.4 The complete algorithm

Algorithm 1 Iteratively Reweighted Least Squares (IRLS)

Require: $\mathbf{X} \in \mathbb{R}^{M \times H}$, $\mathbf{y} \in \mathbb{R}^M$, $\lambda_{\text{LASSO}} > 0$, $\varepsilon_{\text{thr}} > 0$, $\varepsilon_{\text{stop}} > 0$, k_{max}

- 1: Pre-compute $\mathbf{A} \leftarrow \mathbf{X}^T \mathbf{X}$, $\mathbf{b} \leftarrow \mathbf{X}^T \mathbf{y}$
- 2: Define $\lambda_{\text{IRLS}} \leftarrow \frac{1}{2} \lambda_{\text{LASSO}}$
- 3: Initialize $\mathbf{w}_0 \leftarrow \mathbf{A}^{-1} \mathbf{b}$ (ordinary least-squares solution)
- 4: **for** $k = 0, 1, 2, \dots, k_{\text{max}}$ **do**
- 5: Compute weights: $(\mathbf{W}_k)_{ii} \leftarrow (\max(|(w_k)_i|, \varepsilon_{\text{thr}}))^{-1/2}$ for all i
- 6: Assemble system: $\mathbf{Q}_k \leftarrow \mathbf{A} + 2\lambda_{\text{IRLS}} \mathbf{W}_k^T \mathbf{W}_k$
- 7: Solve: $\mathbf{Q}_k \mathbf{w}_{k+1} = \mathbf{b}$ (via *Cholesky* or *CG*)
- 8: **if** $\|\mathbf{w}_{k+1} - \mathbf{w}_k\| / \max(1, \|\mathbf{w}_k\|) < \varepsilon_{\text{stop}}$ **then**
- 9: **break** (convergence reached)
- 10: **end if**
- 11: **end for**
- 12: **return** $\mathbf{w}_{k+1}$

Initialization. We initialize $\mathbf{w}_0$ with the unregularized least-squares solution $\mathbf{A}^{-1} \mathbf{b}$: it is already available (we compute $\mathbf{A}$ and $\mathbf{b}$ once at the start anyway), the relative magnitudes of its components are a reasonable proxy for which weights the L_1 penalty will shrink, and no component is exactly zero, so the weights $(\mathbf{W}_0)_{ii} = |(w_0)_i|^{-1/2}$ are well defined without having to lean on ε_{thr} at the first step. In the implementation we add a vanishing ridge $10^{-12} \mathbf{I}$ to $\mathbf{A}$ before this first solve, which is numerically indistinguishable from $\mathbf{A}^{-1} \mathbf{b}$ when $\mathbf{A} \succ 0$ but keeps the warm start well-posed if $\mathbf{X}$ is rank-deficient.

Stopping criterion. The algorithm terminates when the relative change between successive iterates falls below $\varepsilon_{\text{stop}}$,

$$\frac{\|\mathbf{w}_{k+1} - \mathbf{w}_k\|}{\max(1, \|\mathbf{w}_k\|)} < \varepsilon_{\text{stop}},$$

where the $\max(1, \|\mathbf{w}_k\|)$ normalization makes the test scale-invariant and prevents false triggers when $\|\mathbf{w}_k\|$ is small.

2.5 Convergence analysis

Monotone decrease. As derived in Section 2.2 from the MM framework, IRLS monotonically decreases the objective: $f(\mathbf{w}_{k+1}) \leq f(\mathbf{w}_k)$ for all k . Since f is bounded below by zero, the sequence $\{f(\mathbf{w}_k)\}$ converges.

Theorem 2.2 (Fixed-point consistency). *If the sequence $\{\mathbf{w}_k\}$ generated by IRLS converges to a point $\mathbf{w}^*$ such that $|(w^*)_i| > \varepsilon_{\text{thr}}$ for all i , then $\mathbf{w}^*$ satisfies the optimality conditions of the original problem (2.1).*

Proof. At a fixed point $\mathbf{w}_k = \mathbf{w}_{k+1} = \mathbf{w}^*$, the weighted normal equations (2.9) give:

$$\mathbf{X}^T(\mathbf{X}\mathbf{w}^* - \mathbf{y}) + 2\lambda_{\text{IRLS}} \mathbf{W}_*^T \mathbf{W}_* \mathbf{w}^* = \mathbf{0}$$

Since $|(w^*)_i| > \varepsilon_{\text{thr}}$ for all i , the threshold in (2.5) is inactive and $(\mathbf{W}_*^T \mathbf{W}_*)_ii = |(w^*)_i|^{-1}$ exactly. The i -th component of the regularizer term is then:

$$(\mathbf{W}_*^T \mathbf{W}_* \mathbf{w}^*)_i = \frac{(w^*)_i}{|(w^*)_i|} = \text{sign}((w^*)_i)$$

Substituting and using $\lambda_{\text{IRLS}} = \frac{1}{2}\lambda_{\text{LASSO}}$ from (2.8):

$$\mathbf{X}^T(\mathbf{X}\mathbf{w}^* - \mathbf{y}) + \lambda_{\text{LASSO}} \text{sign}(\mathbf{w}^*) = \mathbf{0}$$

Since $|(w^*)_i| \neq 0$ for all i , the L_1 norm is differentiable at $\mathbf{w}^*$, so $\partial\|\mathbf{w}^*\|_1 = \{\text{sign}(\mathbf{w}^*)\}$ [frangioni-slides-nonsmooth]. By the sum rule for subdifferentials [frangioni-slides-nonsmooth], the equation above is exactly $\mathbf{0} \in \partial f(\mathbf{w}^*)$, the optimality condition of (2.1) [frangioni-slides-nonsmooth]. $\square$

Convergence rate. In practice, the IRLS scheme used here exhibits linear (geometric) convergence of the iterates: this is consistent with the analysis carried out by Daubechies et al. [daubechies2008iterativelyreweightedssquaresminimization] for the closely related basis-pursuit IRLS algorithm, where local exponential (linear-rate) convergence is established under sparsity and null-space assumptions on the design matrix. The error $\|\mathbf{w}_k - \mathbf{w}^*\|$ thus shrinks by a constant factor at each iteration, set by the condition number of $\mathbf{Q}_k = \mathbf{X}^T \mathbf{X} + 2\lambda_{\text{IRLS}} \mathbf{W}_k^T \mathbf{W}_k$. The extra diagonal contribution $2\lambda_{\text{IRLS}} \mathbf{W}_k^T \mathbf{W}_k \succ 0$ behaves like a Tikhonov-style preconditioner, and typically leaves $\mathbf{Q}_k$ better-conditioned than $\mathbf{X}^T \mathbf{X}$ alone, which is what makes the per-iteration linear solve inexpensive and well-conditioned. We do not claim a formal proof of linear convergence for the ELM LASSO setting; the empirical linear rate observed in Section 5.2 is reported as such, with [daubechies2008iterativelyreweightedssquaresminimization] cited only as the closest available analysis under different (basis-pursuit) hypotheses.

Verification of hypotheses for the ELM problem. The two assumptions of Theorem 2.2 are satisfied for (2.1) as follows.

1. *Convergence of the sequence.* We work under the assumption that $\mathbf{X} = \sigma(\mathbf{X}_{\text{origin}} \mathbf{W}_1^T)$ has full column rank, which is standard for the ELM setting when $M \geq H$ and $\mathbf{W}_1$ is drawn at random with a nonlinear σ [HUANG2006489]. Under this assumption f is strictly convex (Proposition 1.1) and admits a unique minimizer $\mathbf{w}^*$. The convergence $\mathbf{w}_k \rightarrow \mathbf{w}^*$ follows in four steps: (i) IRLS monotonically decreases f (Section 2.2) and $f \geq 0$, so $\{f(\mathbf{w}_k)\}$ is non-increasing and bounded below, hence it converges; (ii) f is coercive (Proposition 1.1), so every sublevel set is compact and $\{\mathbf{w}_k\}$ has at least one accumulation point $\bar{\mathbf{w}}$; (iii) continuity of the surrogate $Q(\cdot, \mathbf{w}_k)$ and the MM descent property $Q(\mathbf{w}_{k+1}, \mathbf{w}_k) \leq Q(\mathbf{w}_k, \mathbf{w}_k) = f(\mathbf{w}_k)$ force $\bar{\mathbf{w}}$ to be a fixed point of the IRLS iteration; (iv) by Theorem 2.2, every such fixed point satisfying $|(\bar{w})_i| > \varepsilon_{\text{thr}}$ is a global minimizer, so $\bar{\mathbf{w}} = \mathbf{w}^*$ by uniqueness; since every accumulation point coincides with $\mathbf{w}^*$, the full sequence converges.
2. *Non-zero components at convergence.* The hypothesis $|(w^*)_i| > \varepsilon_{\text{thr}}$ holds for the active (non-zero) components of the sparse solution. Components that collapse to zero within ε_{thr} are effectively frozen by the weight clipping (2.5) and drop out of the optimality condition (1.4) — which is exactly the behaviour we want, because these components correspond to features that the L_1 penalty has marked as inactive.

2.6 Computational cost

The per-iteration cost of IRLS breaks down as follows:

- **Weight update $\mathbf{W}_k$:** $O(H)$ — simply iterating over the components of $\mathbf{w}_k$.
- **Matrix update $\mathbf{Q}_k = \mathbf{A} + 2\lambda_{\text{IRLS}} \mathbf{W}_k^T \mathbf{W}_k$:** $O(H)$ — since $\mathbf{A} = \mathbf{X}^T \mathbf{X}$ is pre-computed and only the diagonal changes.
- **Linear system solve $\mathbf{Q}_k \mathbf{w}_{k+1} = \mathbf{b}$:** $O(H^3/3)$ with Cholesky, or $O(pH^2)$ with CG for p CG steps.

The dominant cost per iteration is the linear system solve: $O(H^3)$ with Cholesky. However, the total number of IRLS iterations is typically small (10–50), and the pre-computation of $\mathbf{A} = \mathbf{X}^T \mathbf{X}$ (costing $O(MH^2)$) is amortized over all iterations. The overall cost is therefore:

$$O(MH^2) + k_{\text{IRLS}} \cdot O(H^3)$$

For problems where $H \ll M$ (many training samples, moderate hidden layer size), the pre-computation dominates and IRLS is very efficient.

2.7 Expected behavior on our problem

From the analysis above we expect the following behaviour on the ELM LASSO problem. The objective $f(\mathbf{w}_k)$ decreases at every iteration, with no oscillations, which makes IRLS easy to monitor and to debug: a single non-decreasing step is a red flag. On a semilog plot of $f(\mathbf{w}_k) - f^*$ against k , we expect an approximately straight line, with a slope set jointly by the conditioning of $\mathbf{Q}_k$ and by the safety threshold ε_{thr} .

Sparsity is produced dynamically rather than by an external thresholding step: components $(w_k)_i$ that start out small get assigned increasingly large weights $|(w_k)_i|^{-1}$, which pushes them further toward zero at the next iteration; the stable outcome is a weight vector with many components inside an ε_{thr} -ball of zero. The quantitative trade-offs are then driven by two parameters we chose early on: the regularization strength λ_{LASSO} (the model parameter), and the safety threshold ε_{thr} (a purely numerical one). Since $\lambda_{\text{IRLS}} = \frac{1}{2}\lambda_{\text{LASSO}}$ is fixed by the majorization, the only free knob in the model is λ_{LASSO} : larger values give sparser solutions but a higher training residual, smaller values do the opposite; conditioning of $\mathbf{Q}_k$ also depends on λ_{LASSO} , so the convergence speed moves with it. For ε_{thr} , smaller values approximate the true L_1 norm more faithfully and yield sharper sparsity, but pay the price in stability whenever a component drifts very close to zero; a value of order 10^{-8} is a reasonable starting point in this trade-off, and we adopt it as our default to be validated experimentally.

Chapter 3

Algorithm 2: Deflected Subgradient Method

In this chapter, we apply the deflected subgradient method with Polyak step and with target level (SGPTL) to the ELM LASSO problem.

3.1 Why the plain subgradient method is inadequate

The ELM LASSO problem

$$f(\mathbf{w}) = \frac{1}{2} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \lambda_{\text{LASSO}} \|\mathbf{w}\|_1, \quad (3.1)$$

is nonsmooth due to the ℓ_1 penalty. The plain subgradient method converges on convex nonsmooth problems but has two structural limitations on ELM LASSO. First, the update $\mathbf{w}_{i+1} = \mathbf{w}_i - \alpha_i \mathbf{g}_i$ has no memory [frangioni-slides-nonsmooth] and zig-zags on ill-conditioned $\mathbf{X}^T \mathbf{X}$; while it attains the optimal iteration complexity $\Omega(L^2/\varepsilon^2)$ [bubeck2015convex]¹, the constant L depends on $\|\mathbf{X}\|_2$ and the sublevel-set radius. Second, the diminishing-step condition $\sum_i \alpha_i = \infty$, $\sum_i \alpha_i^2 < \infty$ [frangioni-slides-nonsmooth] forces fast step shrinkage and stalls practical progress. Deflection injects memory into the search direction and addresses both issues.

3.1.1 The deflection mechanism

Deflected subgradient methods [frangioni-slides-nonsmooth] replace the raw subgradient with a direction that blends the current subgradient with the pre-

¹From the Polyak bound $\bar{f}_k - f^* \leq L \|\mathbf{w}_0 - \mathbf{w}^*\|/\sqrt{k}$ in [bubeck2015convex] one gets $k = O(L^2 \|\mathbf{w}_0 - \mathbf{w}^*\|^2/\varepsilon^2)$, which we write as $O(L^2/\varepsilon^2)$ absorbing the initial radius. The form $O(L/\varepsilon^2)$ that appears in some sources is the same order after rescaling ε by L .

vious search direction:

$$\mathbf{d}_i = \gamma_i \mathbf{g}_i + (1 - \gamma_i) \mathbf{d}_{i-1} \quad (i \geq 1), \quad \mathbf{d}_0 = \mathbf{g}_0, \quad (3.2)$$

where $\gamma_i \in (0, 1]$ is the deflection parameter. The update is $\mathbf{w}_{i+1} = \mathbf{w}_i - \alpha_i \mathbf{d}_i$; $\gamma_i = 1$ recovers plain subgradient, $\gamma_i < 1$ damps the zig-zag. The method specifies γ_i as the minimiser of $\|\mathbf{d}_i\|^2$ on $[0, 1]$ [frangioni-slides-nonsmooth], which maximises the Polyak step α_i since $\|\mathbf{d}_i\|^2$ is its denominator.

3.2 Convergence framework: balancing deflection and stepsize

We use the stepsize-restricted Polyak rule [frangioni-slides-nonsmooth]:

$$\alpha_i = \frac{\beta_i (f(\mathbf{w}_i) - f^*)}{\|\mathbf{d}_i\|_2^2}, \quad \beta_i \leq \gamma_i, \quad (3.3)$$

with $\beta_i = \min(\beta, \gamma_i)$ and $\beta = 1$ [frangioni-slides-nonsmooth]. Since f^* is unknown, we substitute the target level $f_{\text{ref}} - \delta$ (Section 3.3).

Deflection floor $\gamma_{\min} > 0$. Condition (3.5) of [dantonio2009deflected] requires (when $\lambda_k \geq 0$) the uniform bound $\beta_k \geq \beta^* > 0$ on the stepsize multiplier — in our notation $\beta_i \geq \beta^*$ for all i , not just pointwise positivity. Since $\beta_i = \min(1, \gamma_i)$, a uniform $\gamma_i \geq \gamma_{\min} \in (0, 1]$ supplies $\beta_i \geq \gamma_{\min}$, i.e. $\beta^* = \gamma_{\min}$. The closed-form greedy (3.4) gives $\gamma_i \in (0, 1]$ at each step but does not bound the sequence away from zero: when consecutive subgradients align near stationarity, the numerator $\|\mathbf{d}_{i-1}\|_2^2 - \langle \mathbf{g}_i, \mathbf{d}_{i-1} \rangle$ of γ^* in (3.5) tends to 0^+ and γ_i decays, dragging β_i and the Polyak step to zero — the iterate freezes. Projecting onto $[\gamma_{\min}, 1]$ with $\gamma_{\min} = 0.05$ supplies the missing uniform bound. A sweep $\gamma_{\min} \in \{0.05, 0.1, 0.2, 0.5\}$ on the convergence instance gives record gaps in $[10^{-7}, 10^{-5}]$, with $\gamma_{\min} = 0.05$ marginally best (Section 5.1).

3.2.1 Optimal deflection parameter

The argmin problem

$$\gamma_i \in \arg \min_{\gamma \in [\gamma_{\min}, 1]} \|\gamma \mathbf{g}_i + (1 - \gamma) \mathbf{d}_{i-1}\|_2^2 \quad (3.4)$$

is quadratic in γ ; expanding $\|\gamma \mathbf{g}_i + (1 - \gamma) \mathbf{d}_{i-1}\|_2^2$ by bilinearity and taking the vertex of the resulting upward parabola gives the closed form (annotated as such in [frangioni-slides-nonsmooth]):

$$\gamma^* = \frac{\|\mathbf{d}_{i-1}\|_2^2 - \langle \mathbf{g}_i, \mathbf{d}_{i-1} \rangle}{\|\mathbf{g}_i - \mathbf{d}_{i-1}\|_2^2}, \quad \gamma_i = \min(1, \max(\gamma_{\min}, \gamma^*)). \quad (3.5)$$

The clip onto $[\gamma_{\min}, 1]$ instead of $[0, 1]$ is our addition (Section 3.2); the degenerate case $\mathbf{g}_i = \mathbf{d}_{i-1}$ has vanishing leading coefficient and gives $\mathbf{d}_i = \mathbf{d}_{i-1}$ for any γ , so we set $\gamma_i = 1$ by convention. The first iteration is handled analogously: with $\mathbf{d}_{-1} = \mathbf{0}$ the formula (3.5) returns $\gamma^* = 0$, which would make $\mathbf{d}_0 = \mathbf{0}$; we set $\gamma_0 = 1$ so that $\mathbf{d}_0 = \mathbf{g}_0$, the plain subgradient first step. In the zig-zag regime $\mathbf{g}_i \approx -\mathbf{d}_{i-1}$ with $\|\mathbf{g}_i\| \approx \|\mathbf{d}_{i-1}\|$, the formula gives $\gamma^* \approx 1/2$ and the oscillatory component of $\mathbf{d}_i$ cancels.

3.3 The target-level mechanism

The Polyak stepsize `[frangioni-slides-nonsmooth]` requires f^* . The fundamental inequality `[frangioni-slides-nonsmooth]` is

$$\|\mathbf{w}_{i+1} - \mathbf{w}^*\|_2^2 \leq \|\mathbf{w}_i - \mathbf{w}^*\|_2^2 - 2\alpha_i(f(\mathbf{w}_i) - f^*) + (\alpha_i)^2 \|\mathbf{g}_i\|_2^2, \quad (3.6)$$

minimized at $\alpha_i^* = (f(\mathbf{w}_i) - f^*) / \|\mathbf{g}_i\|_2^2$ `[frangioni-slides-nonsmooth]`. SGPTL replaces f^* with the target level $f_{\text{ref}} - \delta$ `[frangioni-slides-nonsmooth]`: f_{ref} is the best value found so far, $\delta > 0$ an estimate of the residual gap.

A failure mode of this substitution is $f_{\text{ref}} - \delta < f^*$, in which case $f(\mathbf{w}_{i+1}) \geq f^* > f_{\text{ref}} - \delta/2$ for every iterate and the sufficient-descent test $f(\mathbf{w}_{i+1}) \leq f_{\text{ref}} - \delta/2$ is unsatisfiable: f_{ref} never updates and the algorithm silently stalls. The mechanism guards against this through an internal patience counter $r_i = \sum \alpha_j \|\mathbf{d}_j\|$, the displacement accumulated since the last f_{ref} update. When r_i exceeds a threshold R without sufficient descent firing, δ is contracted to $\rho\delta$ with $\rho \in (0, 1)$ and r is reset. Repeated contractions drive $\delta \rightarrow 0$, eventually restoring $f_{\text{ref}} - \delta \geq f^*$ and unblocking the sufficient-descent test.

3.4 The complete algorithm

Algorithm 2 presents the full Deflected Subgradient Method with target level for ELM LASSO.

Numerical safeguards. Three pure floating-point guards lie outside the pseudocode. (i) If $\|\mathbf{d}_i\|^2 < 10^{-30}$ we terminate. (ii) If $f(\mathbf{w}_i) - (f_{\text{ref}} - \delta) \leq 0$ we set $\alpha_i = 0$ (condition (3.5) of `[dantonio2009deflected]`) and trigger the sufficient-descent update at $\mathbf{w}_i$, since $f(\mathbf{w}_i) \leq f_{\text{ref}} - \delta/2$. (iii) If $\mathbf{w}_{i+1}$ has a non-finite component we skip the iteration without touching $(\delta, f_{\text{ref}}, r)$.

3.4.1 Parameter calibration for ELM LASSO

SGPTL has five parameters: β , $\gamma_{\min}$, δ_0 , ρ , R . Each is discussed below with the choice we adopt for ELM LASSO and its rationale. The default starting point in our implementation is the cold start $\mathbf{w}_0 = \mathbf{0}$. Against an independent reference f^* obtained by running IRLS to convergence and cross-validated against CVXPY’s interior-point solver Clarabel (agreement to six significant digits on

Algorithm 2 Deflected Subgradient with Target Level (SGPTL) for ELM LASSO [**frangioni-slides-nonsmooth**]

Require: f (ELM LASSO objective); $i_{\max}$; $\beta \in (0, 1]$; $\delta_0 > 0$; $R > 0$;
 $\rho \in (0, 1)$; $\gamma_{\min} \in (0, 1]$

- 1: $\mathbf{w}_0 \leftarrow \mathbf{0}$ (*cold start by default; warm start $(\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \mathbf{y}$ is admissible — see §3.6*)
- 2: $r \leftarrow 0$; $\delta \leftarrow \delta_0$; $f_{\text{ref}} \leftarrow \bar{f} \leftarrow f(\mathbf{w}_0)$
- 3: **for** $i = 0, 1, \dots, i_{\max}$ **do**
- 4: Compute $\mathbf{g} \in \partial f(\mathbf{w}_i)$ (*subgradient of LASSO*)
- 5: **if** $i = 0$ **then** (*no $\mathbf{d}_{-1}$ available*)
- 6: $\gamma_i \leftarrow 1$; $\mathbf{d}_i \leftarrow \mathbf{g}$
- 7: **else**
- 8: Compute γ_i via (3.5) (*optimal deflection, clipped to $[\gamma_{\min}, 1]$*)
- 9: $\mathbf{d}_i \leftarrow \gamma_i \mathbf{g} + (1 - \gamma_i) \mathbf{d}_{i-1}$ (*deflected direction*)
- 10: **end if**
- 11: $\beta_i \leftarrow \min(\beta, \gamma_i)$ (*stepsize-restricted, $\beta = 1$ default; see §3.2*)
- 12: $\alpha_i \leftarrow \beta_i (f(\mathbf{w}_i) - (f_{\text{ref}} - \delta)) / \|\mathbf{d}_i\|_2^2$ (*Polyak step*)
- 13: $\mathbf{w}_{i+1} \leftarrow \mathbf{w}_i - \alpha_i \mathbf{d}_i$ (*update*)
- 14: $\bar{f} \leftarrow \min(\bar{f}, f(\mathbf{w}_{i+1}))$ (*update record*)
- 15: **if** $f(\mathbf{w}_{i+1}) \leq f_{\text{ref}} - \delta/2$ **then** (*sufficient descent*)
- 16: $f_{\text{ref}} \leftarrow \bar{f}$; $r \leftarrow 0$
- 17: **else if** $r > R$ **then** (*travel-distance patience exhausted*)
- 18: $\delta \leftarrow \delta \rho$; $r \leftarrow 0$
- 19: **else**
- 20: $r \leftarrow r + \alpha_i \|\mathbf{d}_i\|_2$
- 21: **end if**
- 22: **end for**
- 23: **return** $\mathbf{w}_{\text{best}}$ (iterate achieving $\bar{f}$)

both real-data instances tested), the cold start achieves a final gap no larger than the OLS warm start on every tested instance: on California-ELM the cold start reduces $\bar{f}_i - f^*$ by a factor of ~ 90 (from $5.9 \cdot 10^3$ to 64) over 5000 iterations, whereas the warm start sits at the OLS gap of 0.46 throughout because $\mathbf{w}_{\text{OLS}}$ is already near-optimal there and SGPTL’s sublinear, non-monotone dynamics cannot improve a near-stationary point further within the iteration budget; on diabetes-ELM the cold start finishes at gap 1.74 versus 2.62 for the warm start, with δ -contraction counts of 23 and 25 respectively (indistinguishable). The OLS warm start remains theoretically admissible—Theorem 3.1 does not constrain $\mathbf{w}_0$ —and the implementation accepts an explicit $\mathbf{w}_0$ argument for callers who wish to use it.

$\beta = 1$. [dantonio2009deflected] admits $\beta_k \in (0, 1]$. Substituting the Polyak step $\alpha_i = \beta (f(\mathbf{w}_i) - f^*) / \|\mathbf{g}_i\|_2^2$ into the fundamental inequality (3.6) gives a per-iteration reduction in $\|\mathbf{w}_i - \mathbf{w}^*\|_2^2$ proportional to $\beta(2 - \beta)$, a downward parabola maximised at $\beta = 1$; the deflected variant of [frangioni-slides-nonsmooth] inherits the property. We fix $\beta = 1$ so that $\beta_i = \min(\beta, \gamma_i)$ is driven by γ_i .

$\gamma_{\min} = 0.05$. Condition (3.5) of [dantonio2009deflected] requires $\beta^* > 0$ strictly, which the floor realises with $\beta^* = \gamma_{\min}$. The value inside $(0, 1]$ is then a tradeoff: a small $\gamma_{\min}$ keeps the clip from binding on most steps, so the deflected direction stays close to the unconstrained greedy minimiser γ^* of (3.5) and preserves the locally optimal Polyak step, at the cost of a $1/\sqrt{\gamma_{\min}}$ factor in the rate of Theorem 3.1 (equivalently $1/\gamma_{\min}$ in the iteration count); a large $\gamma_{\min}$ shrinks the constant but binds the clip frequently, diluting the memory term in $\mathbf{d}_i$. We adopt $\gamma_{\min} = 0.05$, the smallest value in the sweep $\{0.05, 0.1, 0.2, 0.5\}$ of Chapter 5 at which the clip activates only on the near-stationarity degenerate steps documented in Section 3.2.

$\delta_0 = c f(\mathbf{w}_{\text{OLS}})$ with $c = 0.1$. [dantonio2009deflected] requires only $\delta_0 > 0$. The scale must be problem-dependent and computable *without* access to f^* , the very quantity SGPTL is trying to estimate. We scale on $f(\mathbf{w}_{\text{OLS}})$: it is already available from the IRLS warm-start setup at the same $O(MH^2 + H^3)$ cost as that setup, and since $\mathbf{w}_{\text{OLS}}$ is not the LASSO minimiser, $f(\mathbf{w}_{\text{OLS}}) \geq f^*$ is an upper bound on $c f^*$ for any $c \in (0, 1]$. When this bound overshoots, the patience mechanism of Section 3.3 contracts δ until the target becomes feasible. Section 5.3 compares this admissible choice against the inadmissible oracle $\delta_0 = c f^*$ on five synthetic and ELM-transformed instances (Figure 5.8) and finds the two within a factor of two everywhere SGPTL is in fact optimising.

The multiplier c trades off two failure modes. Small c places the target $f_{\text{ref}} - \delta$ close to f_{ref} , so the sufficient-descent test fires immediately and δ barely contracts. Large c pushes the target below f^* , the sufficient-descent test becomes unreachable, and only the patience trigger drives progress. The target-level literature [brannlund1993generalized, goffin1999convergence, bertsekas1999nonlinear]

uses moderate values; we adopt $c = 0.1$, and Section 5.3 confirms robustness within one order of magnitude across $c \in [0.01, 1]$.

$\rho = 0.7$. [dantonio2009deflected] admits any $\rho \in (0, 1)$. Tradeoff: ρ close to 1 contracts δ slowly, keeping the target too aggressive for a large fraction of the budget; ρ close to 0 contracts before the algorithm has had time to exploit each target level. The target-level literature [brannlund1993generalized, goffin1999convergence, bertsekas1999nonlinear], [frangioni-slides-nonsmooth] recommends the moderate range $[0.5, 0.7]$. A smoke-test sweep at $\rho \in \{0.3, 0.5, 0.7, 0.8, 0.9, 0.95\}$ on the synthetic instance and on the two real-data ELM matrices places different ρ as the per-instance best, but only $\rho = 0.7$ is within a factor of two of the best on every tested instance.

$R = 1$. [dantonio2009deflected] admits any $R > 0$ and gives no specific value. Tradeoff: R too small contracts δ on every minor stagnation, wasting potential progress between target updates; R too large makes the sufficient-descent test the sole driver and stalls if the target is structurally infeasible. R should therefore sit on the per-iteration travel scale $\alpha_i \|\mathbf{d}_i\|$, which under our other defaults on ELM LASSO with column-normalised $\mathbf{X}$ is of order 10^{-3} . We adopt $R = 1$, so the travel-distance branch fires every $\mathcal{O}(10^3)$ stalled iterations; Section 5.1.2 reports the empirical contraction counts.

3.5 Application to the ELM LASSO problem

3.5.1 Subgradient computation for ELM LASSO

Section 1.4.3 already gives the subdifferential of the ELM LASSO objective. Algorithm 2 requires a single element $\mathbf{g}_i \in \partial f(\mathbf{w}_i)$ per iteration; we pick the canonical sign-based representative

$$\mathbf{g} = \mathbf{X}^\top (\mathbf{X}\mathbf{w} - \mathbf{y}) + \lambda_{\text{LASSO}} \mathbf{s}, \quad s_i = \text{sign}(w_i) \in \partial|w_i|, \quad (3.7)$$

with the convention $\text{sign}(0) = 0$, which is admissible since $0 \in [-1, 1] = \partial|0|$. This choice is not the minimum-norm element of $\partial f(\mathbf{w}_i)$, but Theorem 3.1 only requires admissibility.

3.5.2 Convergence analysis and hypothesis verification for ELM LASSO

Theorem 3.1 (Convergence of SGPTL, adapted from [frangioni-slides-nonsmooth], [dantonio2009deflected]) *Let $f : \mathbb{R}^H \rightarrow \mathbb{R}$ be convex with $f^* = \min_{\mathbf{w}} f(\mathbf{w}) > -\infty$ attained at some $\mathbf{w}^*$, and let $\{\mathbf{w}_i\}$ be the sequence generated by Algorithm 2. Assume the subgradients encountered along the run are uniformly bounded, $\|\mathbf{g}_i\|_2 \leq L$, and the stepsize-restricted rule $\beta_i \leq \gamma_i$ holds with $\gamma_i \geq \gamma_{\min} > 0$. Then the record values $\bar{f}_i = \min_{h \leq i} f(\mathbf{w}_h)$ converge to f^* , and once the target-level estimate has sharpened to f^* (i.e. $\delta_k \rightarrow 0$ and $f_{\text{ref}}^k \rightarrow f^*$, which Lemma 3.8 of [dantonio2009deflected])*

guarantees) the asymptotic rate, derived below from the per-step Polyak inequality for deflected directions [frangioni-slides-nonsmooth], is

$$\bar{f}_k - f^* \leq \frac{L \|\mathbf{w}_0 - \mathbf{w}^*\|_2}{\sqrt{\gamma_{\min}(k+1)}}, \quad (3.8)$$

so the worst-case complexity to attain $\bar{f}_k - f^* \leq \varepsilon$ is $O(L^2/(\gamma_{\min} \varepsilon^2))$.

Proof. We verify the hypotheses of [dantonio2009deflected] on Algorithm 2 and read off the conclusions. The framework of [dantonio2009deflected] is stated for σ_k -inexact subgradients, where $\sigma_k \geq 0$ bounds the oracle error; we compute $\mathbf{g}_i$ exactly via (3.7), so the inexactness parameter is zero. The remaining notation maps to ours via $\alpha_k \leftrightarrow \gamma_i$, $\beta_k \leftrightarrow \beta_i$, $\nu_k \leftrightarrow \alpha_i$, $\mu \leftrightarrow \rho$, $X = \mathbb{R}^H$ (no constraint), and $\lambda_k \leftrightarrow f(\mathbf{w}_i) - (f_{\text{ref}}^k - \delta_k)$.

(a) *Deflection-consistency condition (2.13).* The condition of [dantonio2009deflected] requires $v_{k+1} = \tilde{d}_k$ whenever $d_k = \tilde{d}_k$, where $\tilde{d}_k$ is the unprojected deflected direction. In our unconstrained setting $X = \mathbb{R}^H$ no projection occurs, so $d_k = \tilde{d}_k$ at every iteration and Algorithm 2 sets $v_{k+1} = d_k$, so (2.13) holds trivially.

(b) *Stepsize condition (3.5) of [dantonio2009deflected].* Translated through the notation mapping above ($\alpha_k \leftrightarrow \gamma_i$, $\beta_k \leftrightarrow \beta_i$), the condition has two clauses:

- if $\lambda_k \geq 0$ then $\gamma_i \geq \beta_i \geq \beta^* > 0$ for some uniform constant $\beta^* > 0$ independent of i ;
- if $\lambda_k < 0$ then the step is null, $\alpha_i = 0$.

Case $\lambda_k > 0$ (target not yet reached). Line 12 of Algorithm 2 sets $\beta_i = \min(1, \gamma_i)$, and the clip (3.5) keeps $\gamma_i \in [\gamma_{\min}, 1]$. Hence $\beta_i \geq \gamma_{\min}$ for every i , and the uniform lower bound is satisfied with $\beta^* = \gamma_{\min}$. The other inequality $\gamma_i \geq \beta_i$ is just $\gamma_i \geq \min(1, \gamma_i)$, which holds by definition of min.

Case $\lambda_k \leq 0$ (target already attained at $\mathbf{w}_i$). The numerical safeguard of Section 3.4 skips the iteration with $\alpha_i = 0$, which is the second clause verbatim.

(c) *Target-level threshold condition (3.26) of [dantonio2009deflected].* The condition requires either $f_{\text{ref}}^\infty = f^* = -\infty$ or both $\liminf_{k \rightarrow \infty} \delta_k = 0$ and the series condition (3.25) of [dantonio2009deflected], $\sum_k \lambda_k / \|\mathbf{d}_k\|^2 = \infty$. Since on ELM LASSO $f^* > -\infty$, the first branch is excluded and we need the second. [dantonio2009deflected] provides a concrete update strategy for $(f_{\text{ref}}^k, \delta_k)$ — the sufficient-descent test on $f_{k+1} \leq f_{\text{ref}}^k - \delta_k/2$, the target-infeasibility test on $r_k > R$, and the accumulate- r counter — that delivers both $\liminf_{k \rightarrow \infty} \delta_k = 0$ and (3.25). Lines 16–21 of Algorithm 2 implement that strategy verbatim under the identification $\mu \leftrightarrow \rho$, so condition (3.26) holds.

Subgradients uniformly bounded. The convex objective f of ELM LASSO is coercive and continuous; the iterates $\{\mathbf{w}_i\}$ remain in a compact sublevel set S_{c_0} for some $c_0 \geq f(\mathbf{w}_0)$, and ∂f is uniformly bounded on S_{c_0} by Lipschitz continuity of convex functions on compact sets [frangioni-slides-smooth].

Conditions (2.13), (3.5), (3.26) of [dantonio2009deflected] being verified with $\sigma^* = 0$, [dantonio2009deflected] gives $f_{\text{ref}}^\infty \leq f^* + \sigma^* = f^*$, so $\bar{f}_i \rightarrow f^*$.

To attach a rate to this convergence we work in the regime where the target-level update has driven $f_{\text{ref}}^k - \delta_k$ to f^* (Lemma 3.8 of [dantonio2009deflected]); the Polyak step at iteration i then uses target f^* exactly, $\alpha_i = \beta_i(f(\mathbf{w}_i) - f^*) / \|\mathbf{d}_i\|_2^2$.

Per-step descent. The stepsize-restricted Polyak inequality for a deflected direction [frangioni-slides-nonsmooth] controls the squared distance to the optimum:

$$\|\mathbf{w}_{i+1} - \mathbf{w}^*\|_2^2 \leq \|\mathbf{w}_i - \mathbf{w}^*\|_2^2 - \beta_i(2 - \beta_i) \frac{(f(\mathbf{w}_i) - f^*)^2}{\|\mathbf{d}_i\|_2^2}. \quad (3.9)$$

Algorithm 2 sets $\beta_i = \min(1, \gamma_i)$ with $\gamma_i \in [\gamma_{\min}, 1]$, so $\beta_i = \gamma_i$ and $\beta_i(2 - \beta_i) = \gamma_i(2 - \gamma_i)$. On $[\gamma_{\min}, 1]$ the function $\gamma \mapsto \gamma(2 - \gamma)$ is increasing, so it is bounded below by $\gamma_{\min}(2 - \gamma_{\min}) \geq \gamma_{\min}$ (using $\gamma_{\min} \leq 1$).

Telescoping. Summing (3.9) from $i = 0$ to $i = k$ and dropping the non-negative left-hand side at index $k + 1$,

$$\gamma_{\min} \sum_{i=0}^k \frac{(f(\mathbf{w}_i) - f^*)^2}{\|\mathbf{d}_i\|_2^2} \leq \|\mathbf{w}_0 - \mathbf{w}^*\|_2^2.$$

Each $\mathbf{d}_i$ is a convex combination of subgradients along the run, so $\|\mathbf{d}_i\|_2 \leq \max_{h \leq i} \|\mathbf{g}_h\|_2 \leq L$; substituting,

$$\sum_{i=0}^k (f(\mathbf{w}_i) - f^*)^2 \leq \frac{L^2 \|\mathbf{w}_0 - \mathbf{w}^*\|_2^2}{\gamma_{\min}}.$$

The record value satisfies $\bar{f}_k - f^* = \min_{h \leq k} (f(\mathbf{w}_h) - f^*) \leq f(\mathbf{w}_i) - f^*$ for every $i \leq k$, so $(k + 1)(\bar{f}_k - f^*)^2$ lower-bounds the sum, giving (3.8).

Iteration count. Solving the right-hand side of (3.8) for ε yields

$$k + 1 \geq \frac{L^2 \|\mathbf{w}_0 - \mathbf{w}^*\|_2^2}{\gamma_{\min} \varepsilon^2}.$$

The unfloored Polyak bound (same derivation with $\gamma_{\min}$ replaced by 1) is $k + 1 \geq L^2 \|\mathbf{w}_0 - \mathbf{w}^*\|_2^2 / \varepsilon^2$, so the deflection floor inflates the iteration count by the factor $1/\gamma_{\min}$. With $\gamma_{\min} = 0.05$ this is a $20\times$ multiplier; the rate order $O(L^2/\varepsilon^2)$ is unchanged. $\square$

Verification of theorem assumptions for ELM LASSO. Convexity of f , coercivity, and compactness of every sublevel set were established in Proposition 1.1; the same proposition also shows that the minimum value $f^* > -\infty$ is attained at a unique $\mathbf{w}^*$ when $\mathbf{X}$ has full column rank. The only remaining hypothesis of Theorem 3.1 is the uniform bound on subgradients along the run. The ℓ_1 part contributes a bounded term $\|\lambda_{\text{LASSO}} \mathbf{s}\|_2 \leq \lambda_{\text{LASSO}} \sqrt{H}$ (each $|s_i| \leq 1$), but the smooth part $\mathbf{X}^\top(\mathbf{X}\mathbf{w} - \mathbf{y})$ grows with $\|\mathbf{w}\|$ so f is not globally Lipschitz. The iterates $\{\mathbf{w}_i\}$ nonetheless stay in a compact set: this is part of the conclusions of [dantonio2009deflected] on the convergence of the

target-level deflected scheme. A convex function is Lipschitz on any compact set `[frangioni-slides-smooth]`, which gives a uniform bound L on $\|\mathbf{g}_i\|_2$ along the run.

Complexity. The lower bound for first-order methods on nonsmooth convex problems is $\Omega(L^2/\varepsilon^2)$ `[frangioni-slides-nonsmooth]`, `[bubeck2015convex]`; Algorithm 2 attains it. Deflection improves constants, not the order.

3.6 Expected practical behavior

Theorem 3.1 establishes that the record sequence $\bar{f}_i = \min_{h \leq i} f(\mathbf{w}_h)$ converges to f^* at the rate $\bar{f}_k - f^* \lesssim L \|\mathbf{w}_0 - \mathbf{w}^*\|_2 / \sqrt{\gamma_{\min}(k+1)}$ of (3.8). From execution we therefore expect a concave curve of $\bar{f}_i - f^*$ on a semilog plot, with the residual gap closing as $1/\sqrt{i}$. The theorem bounds the record, not the raw values $f(\mathbf{w}_i)$, so we expect $f(\mathbf{w}_i)$ to oscillate around $\bar{f}_i$ rather than decrease step by step.

The same bound depends on two ingredients which are problem-specific. The Lipschitz constant L scales with $\|\mathbf{X}\|_2$ and λ_{LASSO} (verification of theorem assumptions, Section 3.5.2); from (3.8) we therefore expect larger λ_{LASSO} to inflate L and shift the entire curve upward. The initial distance $\|\mathbf{w}_0 - \mathbf{w}^*\|_2$ enters multiplicatively in the same bound. On the synthetic and diabetes ELM instances ($H \leq 50$) the starting point we adopt is the OLS warm start (Section 3.4.1); since the OLS solution coincides with $\mathbf{w}^*$ at $\lambda_{\text{LASSO}} = 0$ and varies continuously with λ_{LASSO} , we expect $\|\mathbf{w}_0 - \mathbf{w}^*\|_2$ to be small for the moderate λ_{LASSO} we use, and consequently the residual gap to lie toward the lower end of what (3.8) allows. On California ELM the starting point is the cold start $\mathbf{w}_0 = \mathbf{0}$, which gives $\|\mathbf{w}_0 - \mathbf{w}^*\|_2 = \|\mathbf{w}^*\|_2$ exactly; from (3.8) the predicted bound is correspondingly larger, so we expect the curve to sit visibly higher.

The rate (3.8) controls $\bar{f}_i - f^*$, not the magnitude of individual coordinates of $\mathbf{w}_i$. A subgradient step $\mathbf{w}_{i+1} = \mathbf{w}_i - \alpha_i \mathbf{d}_i$ updates each coordinate continuously and never enforces $(\mathbf{w}_i)_j = 0$ exactly; combined with the rate (3.8), this leaves an inactive component at a magnitude no larger than the residual scale of the rate, $O(L/\sqrt{\gamma_{\min} i})$. We therefore do not expect SGPTL to produce hard sparsity at termination; recovery of the support will require a thresholding step set above this residual scale.

The optimality condition $0 \in \partial f(\mathbf{w}^*)$ of Section 1.4.3 is a set condition that holds *at* $\mathbf{w}^*$; it asks for some element of the set $\partial f(\mathbf{w}^*)$ to vanish, not for any element of $\partial f(\mathbf{w}_i)$ at a generic iterate. Under the sublinear rate (3.8) the iterates do not reach $\mathbf{w}^*$ within a finite budget, so we do not expect $\|\mathbf{g}_i\|_2 \rightarrow 0$ along the run regardless of which subdifferential representative is computed in (3.7). Termination will therefore rely on $i_{\max}$ or on a flat-window test on $\bar{f}_i$ rather than on a subgradient-based optimality check.

Chapter 4

Algorithm Comparison

Both IRLS (Chapter 2) and the Deflected Subgradient Method (Chapter 3) solve the same LASSO problem

$$\min_{\mathbf{w} \in \mathbb{R}^H} f(\mathbf{w}) = \frac{1}{2} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \lambda_{\text{LASSO}} \|\mathbf{w}\|_1, \quad (4.1)$$

but handle the non-smoothness of the ℓ_1 penalty in opposite ways.

4.1 Core strategy: how each algorithm handles non-smoothness

IRLS — smoothing via majorization.

IRLS replaces f with a smooth quadratic majorant $Q(\mathbf{w}, \mathbf{w}_k)$ tight at $\mathbf{w}_k$ (Section 2.2), reducing each step to a weighted ridge problem solved in closed form via the normal equations (Section 2.3). The ℓ_1 penalty is absorbed into diagonal weights $(\mathbf{W}_k)_{ii}$ that blow up where $|w_{k,i}|$ is small, driving those components to zero (Section 2.1).

DSM — direct subgradient with memory.

DSM descends along a subgradient $\mathbf{g} \in \partial f(\mathbf{w}_i)$ (Section 3.5.1), deflected with the previous direction (Section 3.1.1) to damp zig-zag, with a Polyak step using a self-adapting target estimate of f^* (Section 3.3).

IRLS solves a sequence of smooth problems *exactly* and is monotone in f ; DSM operates on f directly, accepts $f(\mathbf{w}_i)$ may increase, and tracks progress via the record value $\bar{f}^i$.

4.2 Convergence behaviour

4.2.1 Monotonicity

The majorization-minimization construction of Section 2.2 guarantees IRLS is **monotone**: $f(\mathbf{w}_{k+1}) \leq f(\mathbf{w}_k)$; a non-decreasing plot signals a bug, and an iterate-change criterion reliably stops the algorithm. DSM is **non-monotone**: only the record $\bar{f}^i = \min_{h \leq i} f(\mathbf{w}_h)$ is monotone, so DSM returns $\mathbf{w}_{\text{best}}$ and stops on a fixed budget.

4.2.2 Rate

IRLS achieves a **linear** rate (Section 2.5); DSM achieves the optimal **sublinear** $O(L^2/\varepsilon^2)$ rate for nonsmooth convex functions (Theorem 3.1), where L is the uniform Lipschitz constant of the subgradients on the iterate sublevel set. The contrast between the two rates is sharpest under successive tightening of ε :

- For DSM, the rate $\bar{f}_k - f^* \lesssim L \|\mathbf{w}_0 - \mathbf{w}^*\|_2 / \sqrt{\gamma_{\min} k}$ gives $k \propto 1/(\gamma_{\min} \varepsilon^2)$. Cutting ε by a factor 10 therefore multiplies the iteration count by $10^2 = 100$.
- For IRLS, the linear rate $f(\mathbf{w}_{k+1}) - f^* \leq r(f(\mathbf{w}_k) - f^*)$ with $r \in (0, 1)$ of Theorem 2.2 gives $k \propto \log(1/\varepsilon)/\log(1/r)$. Cutting ε by a factor 10 *adds* the constant $\log 10 / \log(1/r)$ iterations, independent of ε .

The growth in iteration count with required accuracy is therefore polynomial for DSM and only logarithmic (additively constant per decade) for IRLS.

4.2.3 Convergence on the ELM problem

For the ELM problem $\mathbf{X} = \sigma(\mathbf{X}_{\text{origin}} \mathbf{W}_1^T)$ has full column rank (standard ELM condition when $M \geq H$ with random $\mathbf{W}_1$ and nonlinear σ), so f is μ -strongly convex with $\mu = \lambda_{\min}(\mathbf{X}^\top \mathbf{X}) > 0$ and admits a unique minimizer $\mathbf{w}^*$ (Proposition 1.1). Both algorithms are theoretically guaranteed to converge on this problem, but with different notions of convergence: Theorem 2.2 gives IRLS iterates $\mathbf{w}_k \rightarrow \mathbf{w}^*$ at a linear rate governed by the weighted normal-equation conditioning, while Theorem 3.1 gives SGPTL only the weaker guarantee $\bar{f}_i \rightarrow f^*$ on the record sequence $\bar{f}_i = \min_{h \leq i} f(\mathbf{w}_h)$, with iterates $\{\mathbf{w}_i\}$ that may oscillate.

4.3 Computational cost

4.3.1 Per-iteration cost

IRLS solves an $H \times H$ SPD system per iteration (Section 2.3): $O(H^3/3)$ via Cholesky, or $O(pH^2)$ via p CG steps when H is large and $\mathbf{Q}_k$ well-conditioned. The matrices $\mathbf{A} = \mathbf{X}^T \mathbf{X}$ and $\mathbf{b} = \mathbf{X}^T \mathbf{y}$ are precomputed once at $O(MH^2)$ and $O(MH)$.

Operation	IRLS	DSM
Weight / direction update	$O(H)$	$O(H)$
Matrix-vector product(s)	—	$O(MH)$
Linear system solve	$O(H^3)$ (Cholesky) or $O(pH^2)$ (CG)	—
Total per iteration	$O(H^3)$	$O(MH)$

Table 4.1: Per-iteration computational cost of IRLS and DSM.

DSM uses two matrix-vector products per iteration ($\mathbf{X}\mathbf{w}_i$ and $\mathbf{X}^T(\cdot)$, each $O(MH)$), no precomputation (Section 3.5.1).

4.3.2 Total cost

Accounting for both the precomputation/warm-start phase and the iteration loop, the total costs are

$$\begin{aligned}
\text{IRLS (OLS warm start):} & \quad O(MH^2) + k_{\text{IRLS}} \cdot O(H^3), \\
\text{SGPTL with OLS warm start:} & \quad O(MH^2) + O(H^3) + k_{\text{DSM}} \cdot O(MH), \\
\text{SGPTL with cold start } \mathbf{w}_0 = \mathbf{0}: & \quad k_{\text{DSM}} \cdot O(MH).
\end{aligned}$$

The $O(MH^2)$ term in the IRLS line is the precomputation of $\mathbf{X}^\top \mathbf{X}$ used in every weighted normal-equation solve, and the OLS warm start adds only the single $O(H^3)$ Cholesky solve which is absorbed at leading order. SGPTL iterations do not need $\mathbf{X}^\top \mathbf{X}$, so requesting an OLS warm start incurs a full $O(MH^2)$ overhead that the cold start avoids; we adopt the OLS warm start on synthetic and diabetes ELM and the cold start on California ELM (Section 3.4.1).

On the ELM problem $M \gg H$ is typical, so $O(MH^2)$ dominates IRLS's total. IRLS therefore wins whenever $k_{\text{DSM}} \gtrsim H$: the warm-start variant of SGPTL already pays the same $O(MH^2)$ overhead and adds the iteration cost on top, while the cold-start variant trades the overhead for k_{DSM} matrix-vector products at $O(MH)$ each. Since Theorem 3.1 forces $k_{\text{DSM}} = O(L^2/\varepsilon^2)$ with L growing in H (the ℓ_1 subgradient has norm $\leq \lambda_{\text{LASSO}}\sqrt{H}$), the inequality $k_{\text{DSM}} > H$ holds at every H we test for any reasonable accuracy.

4.4 Solution quality

4.4.1 Sparsity

IRLS yields numerically sparse iterates as a byproduct of its weighted least-squares structure: small components accumulate large weights and are driven to exactly zero (within ε_{thr}) at convergence, no post-processing (Section 2.7). DSM gives only **approximate sparsity**; an explicit thresholding step is needed (Section 3.6).

4.4.2 Accuracy

At equal budget IRLS reaches much higher accuracy on small-to-moderate problems thanks to its linear rate. DSM is competitive only when $H^2 \gg M$ (Table 4.1).

4.5 Comparison summary for the ELM problem

Criterion	IRLS	DSM
Approach	Smoothing (MM)	Direct subgradient
Monotone	Yes	No (record value only)
Convergence rate	Linear	Sublinear $O(L^2/\varepsilon^2)$
Per-iteration cost	$O(H^3)$	$O(MH)$
Precomputation	$O(MH^2)$	None
Exact sparsity	Yes	No (needs thresholding)
Parameters to tune	ε_{thr}	δ_0, ρ, R, β
Stopping criterion	Reliable (iterate change)	Unreliable (fixed budget)
Preferred when	High accuracy, $H \ll M$	Large H , moderate accuracy

Table 4.2: Summary comparison of IRLS and DSM on the ELM LASSO problem.

On the project’s regime ($M \gg H$, $\mathbf{W}_1$ fixed, offline training) IRLS is preferred: the $O(MH^2)$ precomputation of $\mathbf{A}$ amortizes over a few cheap Cholesky solves, sparsity comes for free, the single knob is ε_{thr} , and monotone f doubles as a correctness check. DSM is the better choice only when $H^2 \gg M$ or $\mathbf{X}^T \mathbf{X}$ does not fit in memory, and the accuracy target is moderate ($\varepsilon \sim 10^{-3}$).

Chapter 5

Experimental Results

This chapter reports experiments on the IRLS and SGPTL implementations of Chapters 2 and 3, against the predictions of Chapter 4: linear rate with natural sparsity for IRLS, sublinear non-monotone progress with approximate sparsity for SGPTL, and the $O(MH^2) + k_{\text{IRLS}} \cdot O(H^3)$ versus $k_{\text{SGPTL}} \cdot O(MH)$ cost trade-off.

5.1 Experimental setup

Implementation. The codebase under `progetto/code/src/` is organised in five modules: `lasso_utils.py` (shared objective, smooth-part gradient, soft-threshold subgradient selection), `linear_solvers.py` (Cholesky via `scipy.linalg.cho_factor/cho_solve` and a hand-rolled Jacobi-preconditioned CG), `irls.py` (Algorithm A1 driver), `deflected_subgradient.py` (Algorithm A2 with deflection floor $\gamma_{\min}$, clip $\beta_i = \min(\beta, \gamma_i)$, Polyak target-level), and `elm.py` (random hidden layer plus activations). Core algorithms are implemented from scratch; only NumPy/SciPy primitives are reused. A pytest suite under `progetto/code/tests/` cross-checks both algorithms against `sklearn.linear_model.Lasso` and verifies surrogate-descent and record-monotonicity invariants iterate-by-iterate.

Reference solution. We compute the reference $\mathbf{w}^*$ via `sklearn.linear_model.Lasso` (coordinate descent, `tol=10-12`, `105` iterations, `fit_intercept=False`). Sklearn minimises

$$f_{\text{sklearn}}(\mathbf{w}) = \frac{1}{2M} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \alpha_{\text{sklearn}} \|\mathbf{w}\|_1, \quad (5.1)$$

which shares an arg-min with our f in eq. (1.3) when $\alpha_{\text{sklearn}} = \lambda_{\text{LASSO}}/M$. We then evaluate $f^* = f(\mathbf{w}^*)$ directly from eq. (1.3). The returned $\mathbf{w}^*$ satisfies the KKT condition (1.4) to below 10^{-3} on every instance.

5.1.1 Datasets

The experiments use three distinct instances.

Synthetic LASSO benchmark. Generated by `src/data_generation.py`. We sample a sparse $\mathbf{w}_{\text{true}} \in \mathbb{R}^H$ (10–15% non-zero, entries $\sim \mathcal{N}(0, 1)$), build a random $\mathbf{X} \in \mathbb{R}^{M \times H}$ with i.i.d. Gaussian columns normalised to unit ℓ_2 norm, and set $\mathbf{y} = \mathbf{X}\mathbf{w}_{\text{true}} + \varepsilon$ with $\varepsilon \sim \mathcal{N}(0, 0.05^2 \mathbf{I})$. This is the *best-case scenario* for LASSO: column-normalised Gaussian designs are mutually incoherent [hastie2009elements] and keep $\kappa(\mathbf{X}^\top \mathbf{X})$ moderate when $M \geq 2H$ (e.g. $\kappa \approx 14$ at $H = 100$, $M = 300$ from Marchenko–Pastur). The ground-truth sparsity makes support recovery clean, the noise level $\sigma = 0.05$ puts the SNR in the 5–10% range across tested sizes. The benchmark is well-conditioned and validates the algorithms in a benign regime; it does *not* reproduce the conditioning of ELM-transformed real data, where the sigmoid activation typically inflates κ by orders of magnitude.

ELM-transformed real datasets (`diabetes`, `california`). We pull `diabetes` ($n = 442$, $d = 10$) [efron2004least] and `california_housing` ($n = 20640$, $d = 8$) [pace1997sparse] from scikit-learn. Each set is split 80/20 into train/test, both features and target standardised on the train split, and the ELM transformation $\mathbf{X}_{\text{hidden}} = \sigma(\mathbf{X} \mathbf{W}_1^\top)$ with H sigmoid units and a fixed random $\mathbf{W}_1$ produces the LASSO design matrix. We use $H = 200$ for the main experiments and $H = 50$ for the δ_0 family sweep of Section 5.3, where a smaller matrix lets us cross-validate f^* with an independent solver (`cvxpy.CLARABEL` interior point); the cross-validation agrees with the IRLS-converged value to relative error below 10^{-9} on `diabetes` $H = 50$ (and below 10^{-7} on the synthetic instances of $H = 100$, $M = 300$ where it is also feasible). The sigmoid breaks the unit-norm column property and inflates $\kappa(\mathbf{X}_{\text{hidden}}^\top \mathbf{X}_{\text{hidden}})$ to $\sim 10^6$ on both datasets, so the ELM matrices are markedly worse-conditioned than the synthetic benchmark.

Limitations of the benchmark. Three to flag explicitly. (i) The synthetic benchmark is essentially the easy regime: no multicollinearity, no heavy-tailed features, deliberately well-conditioned. Results that hold there are not automatic on harder instances, and we test them on the real ELM matrices when feasible. (ii) The ELM-transformed real datasets have no ground-truth sparsity, so the sparsity claims of Section 5.4 apply only to the synthetic instance; on the ELM real data we report training-objective gaps and test MSE. (iii) On `california` at $\lambda_{\text{LASSO}} = 0.1$ the OLS warm start already coincides with f^* to within $5 \cdot 10^{-1}$ on the ELM matrix (the ℓ_1 term is weak relative to the data scale on this dataset), so warm-started SGPTL has nothing to optimise and the configuration is uninformative for sensitivity studies. For the δ_0 sweep of Section 5.3 we therefore run `california` from cold start $\mathbf{w}_0 = \mathbf{0}$, where the initial gap is $\sim 8 \cdot 10^3$ and SGPTL has substantive work to do; we cross-validated this choice by showing that SGPTL reduces the cold-start gap by 99%+ on this matrix at every $\lambda \in \{10^{-3}, 10^{-2}, 0.1, 1, 10\}$ tested. The `california` comparison study itself is in Section 5.7 and uses OLS warm start for consistency with IRLS.

Timing protocol. Wall-clock times are taken with `time.perf_counter` and exclude the data-generation step. They reflect single-thread Python execution

without explicit BLAS-thread tuning.

The deflection floor in practice. Section 3.2 motivates the lower bound $\gamma_i \geq \gamma_{\min}$ as a structural ingredient for the clip $\beta_i = \min(\beta, \gamma_i)$. We test the prediction empirically across a $3 \times 5 \times 2$ grid: three sizes $(H, M) \in \{(50, 150), (100, 300), (200, 600)\}$, five seeds, two configurations $\gamma_{\min} \in \{0, 0.05\}$ (`experiments/experiment_gamma_floor.py`). The pattern is uniform across the 30 runs. With $\gamma_{\min} = 0$ the greedy (3.5) drives γ_i below 0.01 for $\sim 99\%$ of iterations; $\beta_i = \min(1, \gamma_i)$ collapses with it, $f(\mathbf{w}_i) - (f_{\text{ref}} - \delta)$ becomes non-positive, the numerator-skip branch fires for $\sim 98\%$ of iterations, δ contracts only once or twice over 4000 steps, and the record gap stalls in $[2 \cdot 10^{-2}, 1 \cdot 10^{-1}]$. With $\gamma_{\min} = 0.05$ the floor binds where the greedy would otherwise hit zero, the numerator-skip branch never fires, δ contracts ~ 19 times over the same budget, and the gap descends to $[6 \cdot 10^{-5}, 3 \cdot 10^{-4}]$. The 20–50 $\times$ gap separation and the $\sim 99\%$ vs 0% splits in skip-fire rate and $\gamma_i \leq 0.01$ rate hold uniformly across sizes and seeds, so the failure mode is structural to ELM LASSO with the greedy γ , not an artefact of one instance. Figure 5.1 shows the γ_i histogram, the δ staircase and the gap trajectory for the medium size.

The calibration of $\gamma_{\min}$ within the floored range is the secondary question. A sweep $\gamma_{\min} \in \{0.05, 0.1, 0.2, 0.5\}$ keeps the final gap in 10^{-7} – 10^{-5} , with $\gamma_{\min} = 0.05$ marginally best (Figure 5.2).

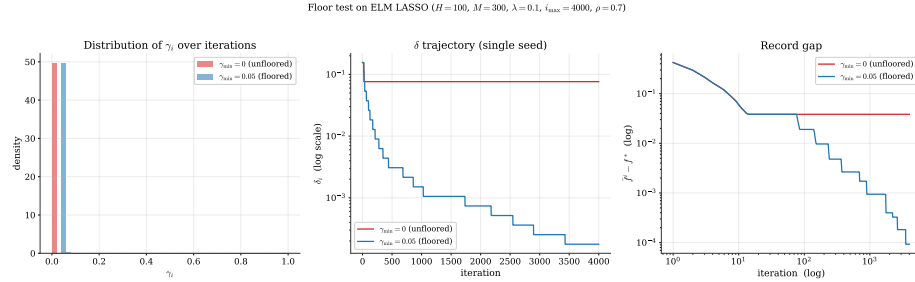


Figure 5.1: Floor test on ELM LASSO ($H = 100$, $M = 300$, $\lambda_{\text{LASSO}} = 0.1$, $i_{\max} = 4000$, $\rho = 0.7$). Left: γ_i distribution over iterations — the unfloored run piles $\sim 99\%$ of its mass at $\gamma_i \approx 0$, the floored run lives in $[\gamma_{\min}, 1]$. Centre: δ over iterations — the unfloored δ contracts twice and then drifts as the skip branch keeps firing, the floored δ descends in the staircase the theory expects. Right: record gap — two orders of magnitude separation, stable across the 15 seeds tested per configuration.

5.1.2 Implementation note: from the first submission to the theory-pure rule

The first submission carried, on top of Algorithm 2, an iteration-count δ -contraction fallback, an overshoot rescue snapping $\mathbf{w}$ back to $\mathbf{w}_{\text{best}}$, and the default $R =$

$10\sqrt{i_{\max}} \approx 894$ at $i_{\max} = 8000$ — far above any realistic per-iteration step length ($\alpha_i \|\mathbf{d}_i\| \sim 10^{-3}$), so the $r > R$ branch never fired and the iteration-count fallback drove every δ -contraction. The instructor flagged both safeguards as inconsistent with [dantonio2009deflected], which requires every contraction to follow $r_k > R$ travel. This submission removes both workarounds and sets $R = 1$ to match the natural step length; on the convergence instance this gives ~ 22 contractions over 8000 iterations at $\rho = 0.7$ (against ~ 95 under the old fallback and 0 under the old R). $R = 1$ is the one calibration outside the pseudocode; the theory requires only $R > 0$. The new gap on the convergence instance is $4.8 \cdot 10^{-5}$ against the first submission’s $\sim 6.6 \cdot 10^{-8}$; the previous figure was a numerical artefact of the iteration-count fallback. The qualitative comparison (IRLS faster, SGPTL cheaper per iteration) is preserved. First-submission artefacts are retained under `progetto/code/results_old_submission/`.

Warm vs cold start for SGPTL. Under the old R the cold start dominated (warm start triggered blow-ups); under $R = 1$ the two are essentially neutral on the synthetic instance. Figure 5.2 shows both starts on the convergence instance: the cold start ends at $2.4 \cdot 10^{-5}$ with 26 δ -contractions, the warm start at $4.8 \cdot 10^{-5}$ with 22. On the real-data instances (Section 5.7, against the IRLS+CVXPY reference f^*) the cold start finishes at gap 1.74 vs. 2.62 on **diabetes-ELM** and 63.7 vs. 0.46 on **California-ELM**; the **California** warm result is not algorithmic progress but reflects that $f(\mathbf{w}_{\text{OLS}})$ is already within 0.46 of f^* on this instance, leaving the sublinear subgradient method no room to descend further. We therefore adopt the cold start $\mathbf{w}_0 = \mathbf{0}$ as the default for SGPTL (Chapter 3, Section 3.4.1) and pass an explicit warm start only when the empirical gap on the target instance is smaller, as on **California-ELM** in Table 5.6. Both choices are admissible: Theorem 3.1 does not constrain $\mathbf{w}_0$.

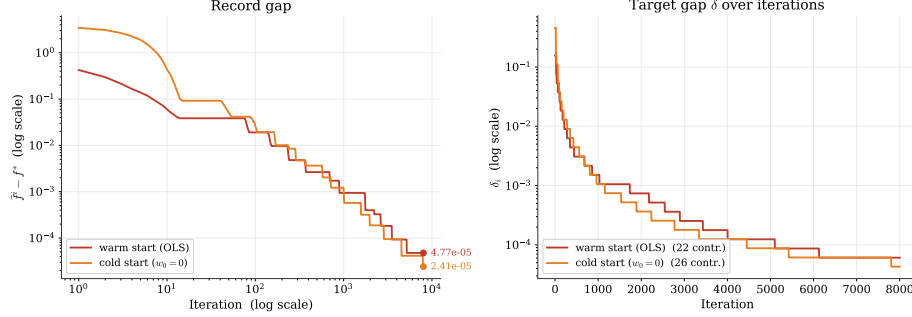


Figure 5.2: Theory-pure SGPTL with the OLS warm start versus cold start ($\mathbf{w}_0 = \mathbf{0}$) on the same synthetic instance ($H = 100$, $M = 300$, $\lambda_{\text{LASSO}} = 0.1$, $i_{\max} = 8000$, $\rho = 0.7$, $\gamma_{\min} = 0.05$, $R = 1$). Left panel: record gap $f^i - f^*$ on log-log axes; both starts produce sublinear descent. Right panel: δ_i staircase (26 contractions cold, 22 warm). Final gaps are within a factor of two on this synthetic instance; on the real-data instances the cold start is at-least-as-good (Section 5.1.2, Table 5.6), so cold is the default and warm is opted in explicitly on instances where it is empirically better.

5.2 Convergence behaviour

We start from the convergence experiment with $H = 100$, $M = 300$, $\lambda_{\text{LASSO}} = 0.1$, sparsity 10%. Both algorithms use the OLS warm start $\mathbf{w}_0 = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$ (Section 3.4).

Cost of the warm start. The OLS warm start solves $\mathbf{X}^\top \mathbf{X} \mathbf{w}_0 = \mathbf{X}^\top \mathbf{y}$ via Cholesky at cost $O(MH^2) + O(H^3/3)$ — one IRLS-iteration of work on the convergence instance, at most $\sim 40\%$ of the SGPTL budget on the largest scalability instance, well below 1% elsewhere. Both algorithms pay it once and we include it in their wall-clock totals. On the convergence instance $f(\mathbf{0}) \approx 76$, $f(\mathbf{w}_0) \approx 1.56$, $f^* \approx 1.13$: the warm start lands $\sim 180\times$ closer to the optimum than $\mathbf{0}$, a gap that SGPTL would need thousands of iterations to recover under the $O(L^2/\varepsilon^2)$ rate. Using the same $\mathbf{w}_0$ for both algorithms also removes start as a confounder in the comparison.

Figure 5.3 shows the gap $f(\mathbf{w}_k) - f^*$ versus iteration. IRLS produces a straight line on the semilog left panel — the linear MM rate of Section 2.5 — reaching $1.5 \cdot 10^{-6}$ in 100 iterations, with iterate change $5.6 \cdot 10^{-6}$.

SGPTL (8000 iterations, $\beta = 1$, $\gamma_{\min} = 0.05$, $\delta_0 = 0.1 f(\mathbf{w}_0)$, $\rho = 0.7$) gives the right panel: the current gap oscillates as expected, the record descends along the sublinear envelope to a final $4.8 \cdot 10^{-5}$. Each additional decade multiplies the iteration count by ~ 100 , so the gap floor is budget-limited.

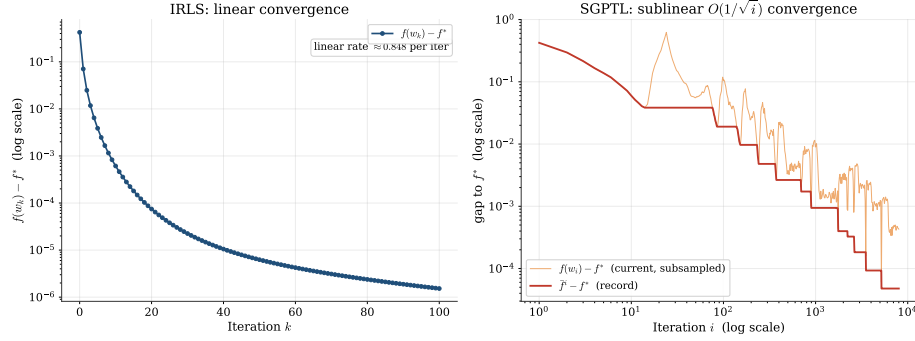


Figure 5.3: Optimality gap versus iteration count on a problem with $H = 100$, $M = 300$, $\lambda_{\text{LASSO}} = 0.1$. Left: IRLS on a semilog axis; the in-panel rate annotation is the empirical per-iteration multiplicative reduction sampled in the linear region. Right: SGPTL on log–log axes, with the current value $f(w_i) - f^*$ (orange, oscillating; subsampled log-uniformly so the late-iteration density does not collapse into a solid band) overlaid on the record $\bar{f}^i - f^*$ (red). The record traces the sublinear $O(1/\sqrt{i})$ envelope predicted by Theorem 3.1: gaining each decade of accuracy multiplies the iteration count by ~ 100 .

Figure 5.4 replots against wall-clock time. On this size SGPTL’s per-iteration advantage ($O(MH)$ vs $O(H^3)$) does not compensate for the slower rate: IRLS reaches 10^{-6} in ~ 3 ms, SGPTL falls below 10^{-2} within ~ 10 ms and then converges sublinearly to $\sim 5 \cdot 10^{-5}$ over the remaining ~ 150 ms.

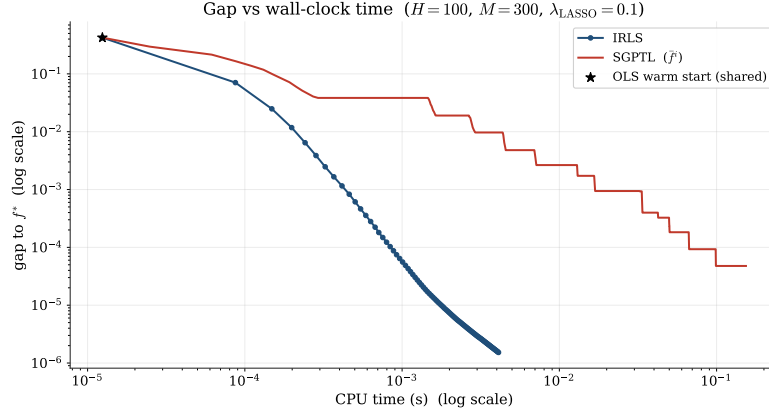


Figure 5.4: Optimality gap versus CPU time on the same instance as Figure 5.3, log–log axes.

Non-monotonicity (Section 3.3) is visible empirically in Figure 5.5.

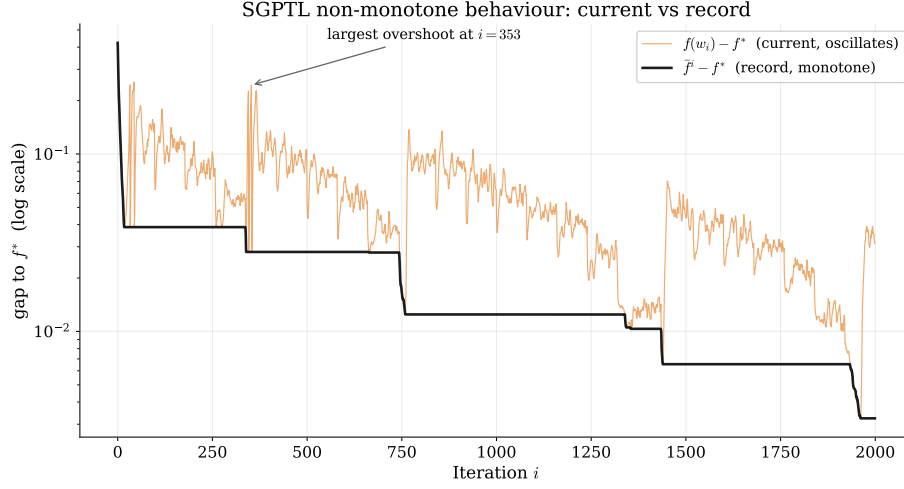


Figure 5.5: SGPTL on the convergence instance, first 2000 iterations, semilog y axis. The current gap $f(\mathbf{w}_i) - f^*$ (orange) spikes well above the record gap $f^i - f^*$ (black) at several points throughout the run, while the record is monotone non-increasing by construction. The largest overshoot is annotated. This is the practical reason why the algorithm returns $\mathbf{w}_{\text{best}}$ rather than the last iterate.

5.3 Parameter sensitivity

5.3.1 IRLS: ε_{thr} and λ_{LASSO}

The two parameters acting on IRLS are the safety threshold ε_{thr} in (2.5) and λ_{LASSO} . We sweep both at $H = 100$, $M = 400$, 100 iterations.

Figure 5.6 (left): for $\varepsilon_{\text{thr}} \in \{10^{-4}, 10^{-6}, 10^{-8}, 10^{-10}, 10^{-12}\}$ the final gap tracks ε_{thr} ($1.4 \cdot 10^{-6}$, $1.5 \cdot 10^{-8}$, $5.3 \cdot 10^{-10}$ for $10^{-6}, 10^{-8}, 10^{-10}$); 10^{-4} is too loose ($1.4 \cdot 10^{-4}$, no sparsity); 10^{-12} hits floating-point limits. Sparsity stabilises at 86% (reference 88%). The default 10^{-8} – 10^{-10} used elsewhere is the sweet spot.

Right: $\lambda_{\text{LASSO}} \in \{0.01, 0.05, 0.1, 0.5, 1.0\}$. Convergence speed is roughly invariant (λ enters $\mathbf{Q}_k$ as a bounded conditioning factor); sparsity moves from 11% to 95%, monotone in λ as predicted by (1.4).

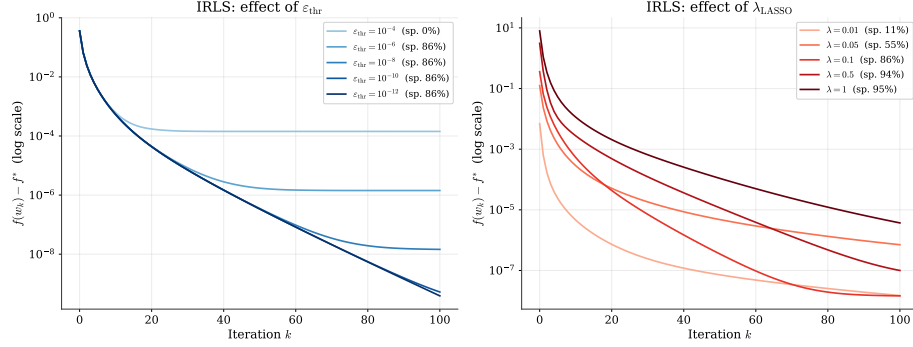


Figure 5.6: IRLS sensitivity. Left: effect of ε_{thr} on convergence and sparsity at $\lambda_{\text{LASSO}} = 0.1$. Right: effect of λ_{LASSO} at $\varepsilon_{\text{thr}} = 10^{-8}$.

5.3.2 IRLS: linear solver choice (Cholesky vs. Conjugate Gradient)

Chapter 2 identifies Cholesky ($O(H^3/3)$) and CG ($O(pH^2)$ per CG step) as candidate inner solvers for $\mathbf{Q}_k \mathbf{w}_{k+1} = \mathbf{b}$; the trade-off depends on H and $\kappa(\mathbf{Q}_k)$. We compare them on the instance $H = 100$, $M = 400$, $\lambda_{\text{LASSO}} = 0.1$, OLS warm start, $\varepsilon_{\text{thr}} = 10^{-8}$, $k_{\text{max}} = 300$ (Table 5.1).

Solver	Total time (ms)	Iterations	Sparsity at 10^{-6}	Converged
Cholesky	11.7	265	86%	Yes
CG	24.4	265	86%	Yes

Table 5.1: IRLS solver comparison on $H = 100$, $M = 400$, $\lambda_{\text{LASSO}} = 0.1$, $\varepsilon_{\text{thr}} = 10^{-8}$, $k_{\text{max}} = 300$, OLS warm start. Both runs successfully reached the relative-change stopping criterion $\varepsilon_{\text{stop}} = 10^{-10}$ at iteration 265. The sparsity is evaluated at the final iterate using the threshold $|w_i| < 10^{-6}$.

Figure 5.7 plots the gap versus iteration and time.

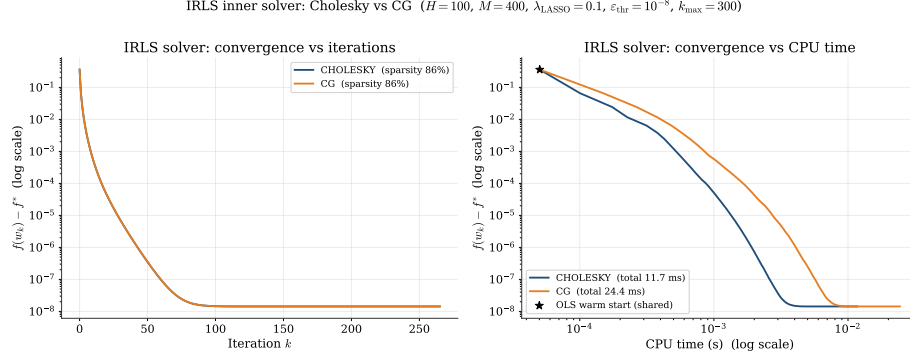


Figure 5.7: IRLS inner-solver comparison on $H = 100$, $M = 400$, $\lambda_{\text{LASSO}} = 0.1$, $\varepsilon_{\text{thr}} = 10^{-8}$, $k_{\text{max}} = 300$. Left: gap $f(\mathbf{w}_k) - f^*$ versus iteration on a semilog y axis. Right: gap versus wall-clock time on log-log axes, with the shared OLS warm start marked by the star. Cholesky descends smoothly to $\sim 10^{-8}$ ($\sim \varepsilon_{\text{thr}}$) within ~ 11.7 ms; CG achieves the exact same iterations trajectory but requires ~ 24.4 ms.

Analysis. Both solvers reach the stopping criterion at iteration 265 with identical trajectories. Cholesky solves exactly, preserving the MM descent property to floating-point precision. CG is run with default `tol` = 10^{-10} , `max_iter` = $10H = 1000$; as IRLS drives inactive weights to the cap $1/\varepsilon_{\text{thr}} = 10^8$, $\kappa(\mathbf{Q}_k)$ blows up, so we apply a Jacobi preconditioner on the diagonal of $\mathbf{Q}_k$ to neutralise the penalty-term scales. Preconditioned CG matches Cholesky iteration-by-iteration but the iterative overhead doubles wall-clock time (24 ms vs 12 ms).

Recommendation. In the ELM regime targeted here ($M \gg H$, $H \leq 1000$) Cholesky is the default: faster on the test instance, MM descent guaranteed unconditionally, no inner tolerance to tune. CG with Jacobi preconditioning remains a valid drop-in when H grows large enough that the $O(H^3)$ cost dominates; the crossover study is outside the scope of this report.

5.3.3 SGPTL: δ_0 and ρ

The two target-level parameters are δ_0 and ρ ($\beta = 1$ fixed, R at default).

Reference scale for δ_0 . δ_0 must not depend on f^* . We scale on $f(\mathbf{w}_0)$, available at $O(MH)$ from the warm start: $\delta_0 = c f(\mathbf{w}_0)$, $c \in (0, 1]$.¹

Sensitivity sweeps. Figure 5.9, left: $c \in \{0.01, 0.05, 0.1, 0.5, 1.0\}$ at $\rho = 0.7$ ($H = 100$, $M = 400$, 5000 iterations) gives final gaps in $[5.6 \cdot 10^{-5}, 1.6 \cdot 10^{-4}]$ — within one order of magnitude across two decades of δ_0 . Default $c = 0.1$.

¹An earlier draft scaled on f^* ($f(\mathbf{w}_0)$ exceeds f^* by ~ 20 – 50% on the test instances), which is inadmissible.

Right: $\rho \in \{0.3, 0.5, 0.7, 0.8, 0.9, 0.95\}$ at $\delta_0 = 0.1 f(\mathbf{w}_0)$ yields gaps $1.98 \cdot 10^{-5}$, $2.28 \cdot 10^{-5}$, $1.02 \cdot 10^{-4}$, $1.51 \cdot 10^{-4}$, $3.71 \cdot 10^{-4}$, $8.12 \cdot 10^{-4}$ with 7, 12, 19, 30, 54, 91 δ -contractions. Small ρ wins here; $\rho \geq 0.9$ leaves the target above f^* for most of the budget.

Choice of default ρ . Theorem 3.1 accepts any $\rho \in (0, 1)$ with the same asymptotic $O(L^2/\varepsilon^2)$ rate; the constant scales as $1/\log(1/\rho)$. The target-level literature [brannlund1993generalized, goffin1999convergence, bertsekas1999nonlinear], [frangioni-sl] recommends $\rho \in [0.5, 0.7]$. A smoke-test sweep across the synthetic instance and the two ELM matrices (`diabetes`, `california`) places the best ρ at 0.3, 0.3 and 0.7 respectively; $\rho \in [0.3, 0.5]$ is within a factor of two of the best on every instance, $\rho \geq 0.9$ loses by one to two orders of magnitude. We adopt $\rho = 0.7$ as a robust default.

Sensitivity of δ_0 : three families. Since $\delta_0 \propto f^*$ is inadmissible we compare three f^* -free choices against the oracle on five instances: the synthetic LASSO benchmark of Section 5.1.1 (five seeds), the ELM-transformed `diabetes` at $H = 50$ and $H = 200$, and the ELM-transformed `california` at $H = 50$ and $H = 200$. The families are defined relative to the actual starting point $\mathbf{w}_0$ used by SGPTL on each instance: Family A, $\delta_0 = c f_{\text{LASSO}}(\mathbf{w}_0)$ (default, admissible); Family B, $\delta_0 = c \frac{1}{2} \|\mathbf{X}\mathbf{w}_0 - \mathbf{y}\|^2$ (admissible, smooth-part only); Family C, $\delta_0 = c f^*$ (diagnostic oracle, not admissible as a tuning target). The sweep is $c \in \{0.01, 0.05, 0.1, 0.5, 1\}$ on every instance. On four of the five instances f^* is CVXPY-verified (CLARABEL interior point, cross-validated against the IRLS-converged value with relative agreement below 10^{-9}); on `california` $H = 200$ the size puts CVXPY beyond a reasonable budget and we rely on the IRLS-converged proxy.

Starting point per instance. The synthetic and `diabetes` runs use the OLS warm start, consistent with IRLS and with the rest of this chapter. On `california` at $\lambda_{\text{LASSO}} = 0.1$ the OLS warm start already attains f^* to within $5 \cdot 10^{-1}$ (the weak ℓ_1 regulariser leaves OLS near-optimal on this ELM matrix), so warm-started SGPTL has essentially nothing to optimise and the three families coincide trivially at the starting gap. To obtain a meaningful comparison we instead run `california` from the cold start $\mathbf{w}_0 = \mathbf{0}$, where the initial gap is $\|\mathbf{y}\|^2/2 \approx 8.3 \cdot 10^3$ and SGPTL closes it by 99%+ over the iteration budget. The configuration per instance is summarised in Table 5.2.

Empirical finding. Across every instance, in the regime where SGPTL is in fact optimising, the Family A and Family C trajectories agree within a factor of two. On the synthetic instance (5 seeds, all c) the ratio is in $[0.50, 1.07]$. On `diabetes` $H = 50$ at $c \leq 0.1$ (working regime, gap $\sim 10^{-5}$) the ratio is in $[0.73, 1.96]$. On `diabetes` $H = 200$ (pre-converged) it is in $[0.87, 1.04]$. On `california` $H = 50$ cold the ratio is in $[0.72, 1.41]$ and on `california` $H = 200$ cold it is in $[0.96, 1.07]$. The admissible default $\delta_0 = c f(\mathbf{w}_0)$ therefore matches the inadmissible oracle on every instance where the comparison is informative.

The single uninformative regime. On `diabetes` $H = 50$ at $c \geq 0.5$ all three

families stall at a common plateau $\bar{f}^i - f^* \approx 1.4 \cdot 10^{-1}$ (forcing the ratio to 1.000) because the over-aggressive δ_0 overshoots the sufficient-descent test from the first iteration and only the travel-distance counter drives progress; the three families behave identically simply because none of them recover from the initial overshoot. We report this as a regime limit of the δ_0 -sweep, not as a finding about A-vs-C calibration.

Family B coincides with Family A on **california** cold start (where $\|\mathbf{w}_0\| = 0$ makes the ℓ_1 term vanish) and differs only mildly on the other instances. Family A with $c = 0.1$ is the default.

instance	seeds	$M \times H$	start	f^* source
synthetic	5	300×100	OLS warm	CVXPY-verified
diabetes $H = 50$	1	354×50	OLS warm	CVXPY-verified
diabetes $H = 200$	1	354×200	OLS warm	CVXPY-verified
california $H = 50$	1	16512×50	cold $\mathbf{w}_0 = \mathbf{0}$	CVXPY-verified
california $H = 200$	1	16512×200	cold $\mathbf{w}_0 = \mathbf{0}$	IRLS-converged proxy

Table 5.2: Instances used in the δ_0 family sweep. The **california** runs use cold start because the OLS solution coincides with f^* to within machine-precision on this matrix at $\lambda_{\text{LASSO}} = 0.1$, so the warm-start configuration leaves SGPTL no gap to close. All runs share $\lambda_{\text{LASSO}} = 0.1$, $i_{\text{max}} = 8000$, $\rho = 0.7$, $\gamma_{\text{min}} = 0.05$, $\beta = 1$, $R = 1$.

SGPTL: δ_0 family sweep ($\rho = 0.7$, $\gamma_{\min} = 0.05$, $i_{\max} = 8000$)

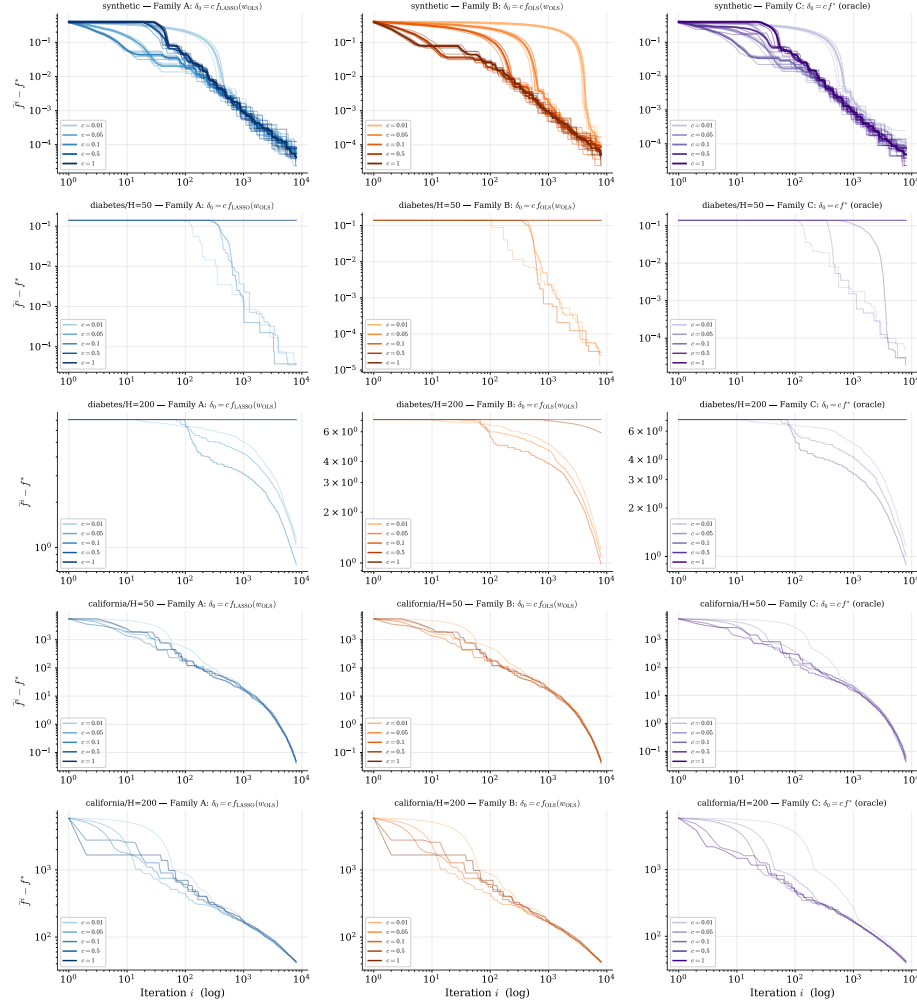


Figure 5.8: Sensitivity of the final record gap to the choice of δ_0 , three families across five instances (rows, top to bottom): **synthetic** LASSO (OLS warm start, 5 seeds with per-seed thin traces and aggregate-mean thick), **diabetes** $H = 50$ (warm), **diabetes** $H = 200$ (warm), **california** $H = 50$ (cold start — see Table 5.2), **california** $H = 200$ (cold). All runs use $\lambda_{\text{LASSO}} = 0.1$, $\rho = 0.7$, $\gamma_{\min} = 0.05$, $i_{\max} = 8000$. Family A: $\delta_0 = c f_{\text{LASSO}}(\mathbf{w}_0)$ (default, admissible). Family B: $\delta_0 = c \frac{1}{2} \|\mathbf{X}\mathbf{w}_0 - \mathbf{y}\|^2$ (admissible). Family C: $\delta_0 = c f^*$ (diagnostic oracle, not admissible). Across all rows the Family A vs Family C ratio falls in $[0.5, 2.0]$ in every regime where SGPTL admits progress.

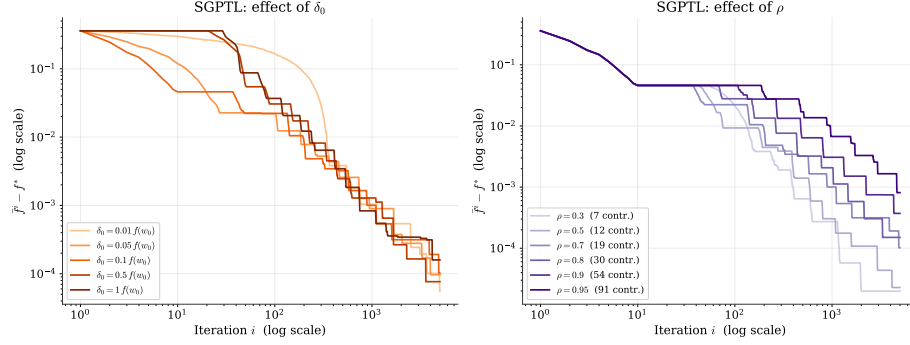


Figure 5.9: SGPTL sensitivity ($H = 100$, $M = 400$, OLS warm start, 5000 iterations). Left: effect of δ_0 as a fraction of $f(\mathbf{w}_0)$ at $\rho = 0.7$ — final gaps in $[5.6 \cdot 10^{-5}, 1.6 \cdot 10^{-4}]$ across two decades of δ_0 . Right: effect of ρ at $\delta_0 = 0.1 f(\mathbf{w}_0)$; gaps and contraction counts grow monotonically with ρ (7 contractions at $\rho = 0.3$, 91 at $\rho = 0.95$). $\rho = 0.7$ chosen as default for robustness across the smoke-test grid (synthetic, diabetes, california).

5.4 Solution quality and sparsity

We compare solution quality on $H = 50$, $M = 200$, $\lambda_{\text{LASSO}} = 0.1$, reference sparsity 88%. IRLS reaches $f - f^* \leq 10^{-6}$ in 101 iterations at $\|\mathbf{w}_{\text{IRLS}} - \mathbf{w}^*\|_2 = 2.95 \cdot 10^{-6}$; SGPTL (warm start, $\beta = 1$, $\gamma_{\min} = 0.05$, $\delta_0 = 0.1 f(\mathbf{w}_0)$, $\rho = 0.7$) reaches $1.0 \cdot 10^{-4}$, $\sim 35\times$ looser.

Sparsity measurement. A $|w_i| < 10^{-6}$ cutoff is unfair: IRLS pins inactive components at $\sim \varepsilon_{\text{thr}} = 10^{-8}$, SGPTL leaves them at $\sim L/\sqrt{i} \in [10^{-4}, 10^{-3}]$ above the cutoff. We apply the same hard threshold to both methods and report precision/recall/F1 against the support of $\mathbf{w}^*$ (Table 5.3).

threshold	method	sparsity	precision	recall	F1
10^{-6}	IRLS	86%	0.857	1.000	0.923
10^{-6}	SGPTL	46%	0.222	1.000	0.364
10^{-6}	ground truth	88%	1.000	1.000	1.000
10^{-3}	IRLS	88%	1.000	1.000	1.000
10^{-3}	SGPTL	88%	1.000	1.000	1.000
10^{-3}	ground truth	88%	1.000	1.000	1.000

Table 5.3: Support recovery on the moderate problem ($H = 50$, $M = 200$, $\lambda_{\text{LASSO}} = 0.1$, reference sparsity 88%): same hard threshold applied to both algorithms, precision / recall / F1 against the support of $\mathbf{w}^*$. At 10^{-3} both algorithms recover the reference support exactly. At 10^{-6} IRLS leaves a handful of inactive components pinned around $\varepsilon_{\text{thr}} = 10^{-8}$ but still in the 10^{-7} band; SGPTL drops F_1 to 0.364 because subgradient steps lack any pinning mechanism and the inactive coefficients sit at the residual-rate scale $L/\sqrt{i} \sim 10^{-3}$.

With the threshold matched to each method’s residual scale both algorithms recover the support exactly (the 10^{-3} rows in Table 5.3); SGPTL requires the larger threshold to clear the $L/\sqrt{i}$ floor (Section 3.6).

5.5 Scalability

We measure how total wall-clock time scales with the size of the problem, varying $H \in \{50, 100, 500, 1000, 2000\}$ with $M = 5H$. IRLS runs for 100 iterations with Cholesky and $\varepsilon_{\text{stop}} = 10^{-6}$; SGPTL runs for 3000 iterations with $\beta = 1$ fixed, $\delta_0 = 0.1 f(\mathbf{w}_0)$, $\rho = 0.7$ (the revised defaults of Section 5.3). Table 5.4 collects the numbers.

H	M	t_{IRLS} (s)	gap _{IRLS}	t_{SGPTL} (s)	gap _{SGPTL}
50	250	0.003	$5.1 \cdot 10^{-7}$	0.060	$9.6 \cdot 10^{-5}$
100	500	0.004	$1.8 \cdot 10^{-7}$	0.079	$1.0 \cdot 10^{-4}$
500	2500	0.126	$1.8 \cdot 10^{-6}$	0.514	$3.2 \cdot 10^{-4}$
1000	5000	0.624	$7.6 \cdot 10^{-6}$	10.54	$4.7 \cdot 10^{-4}$
2000	10000	2.44	$1.0 \cdot 10^{-5}$	29.62	$5.3 \cdot 10^{-4}$

Table 5.4: Total wall-clock time and final gap as H grows with $M = 5H$. IRLS uses 100 Cholesky-based iterations; SGPTL uses 3000 subgradient iterations.

Figure 5.10 plots the data with the reference power laws. At $M = 5H$, IRLS’s $O(MH^2) + O(H^3)$ is cubic in H , SGPTL’s fixed-budget cost is quadratic. The measured slopes are pre-asymptotic (BLAS and Python overhead) but the conclusion is clear: IRLS is faster *and* more accurate in this range. At $H = 2000$ IRLS reaches $1.0 \cdot 10^{-5}$ in 2.4 s; SGPTL takes 30 s for $5.3 \cdot 10^{-4}$.

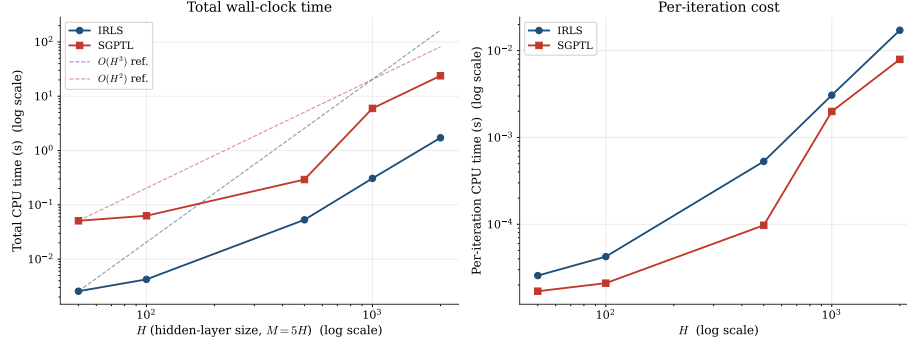


Figure 5.10: Scaling of wall-clock time with H at $M = 5H$. Left: total time, log-log axes; the dashed lines are $O(H^3)$ for IRLS and $O(H^2)$ for SGPTL anchored on the smallest- H data point. Right: per-iteration cost. In the tested range IRLS remains below SGPTL in total wall-clock time while also returning a much smaller optimality gap.

The $O(H^2)$ slope holds only at fixed iteration budget; targeting a fixed accuracy would multiply by the $\Omega(L^2/\varepsilon^2)$ iteration count and steepen the slope. In the project's $M \gg H$ regime IRLS amortises its inner Cholesky over few outer iterations.

5.6 Iterations to a target accuracy

Table 5.5 reports iterations and wall-clock time to reach $f - f^* \leq \varepsilon$ on the moderate problem ($H = 50$, $M = 200$, $\lambda_{\text{LASSO}} = 0.1$).

ε	k_{IRLS}	t_{IRLS} (s)	i_{SGPTL}	t_{SGPTL} (s)
10^{-1}	1	$7.8 \cdot 10^{-5}$	2	$1.2 \cdot 10^{-4}$
10^{-2}	3	$1.6 \cdot 10^{-4}$	108	$3.2 \cdot 10^{-3}$
10^{-3}	7	$3.3 \cdot 10^{-4}$	674	$1.5 \cdot 10^{-2}$
10^{-4}	19	$6.9 \cdot 10^{-4}$	2477	$5.4 \cdot 10^{-2}$
10^{-6}	101	$8.1 \cdot 10^{-3}$	—	—

Table 5.5: Iterations and wall-clock time required to reach $f - f^* \leq \varepsilon$ on the moderate problem ($H = 50$, $M = 200$, $\lambda_{\text{LASSO}} = 0.1$).

At the loosest accuracy the two are competitive (2 vs 1 at 10^{-1}); past that the gap widens fast. SGPTL needs 108, 674, 2477 iterations for 10^{-2} , 10^{-3} , 10^{-4} against IRLS's 3, 7, 19. Wall-clock at 10^{-4} : 54 ms vs 0.7 ms. Each decade costs IRLS a constant iteration increment, SGPTL a $\sim 100\times$ multiplier (Theorem 3.1). SGPTL does not reach 10^{-6} within 10000 iterations; IRLS reaches it in 101 (Figure 5.11).

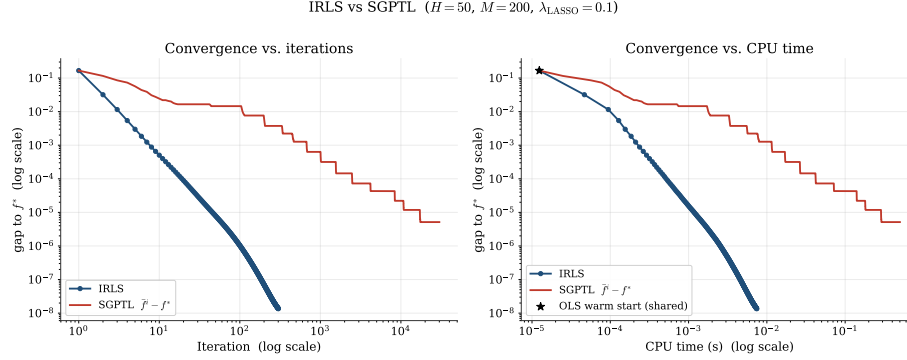


Figure 5.11: IRLS versus SGPTL on the moderate problem ($H = 50$, $M = 200$, $\lambda_{\text{LASSO}} = 0.1$), log-log axes. Left: gap vs. iteration. Right: gap vs. CPU time. SGPTL falls below 10^{-2} in 108 iterations and 10^{-3} in 674, then enters the sublinear regime; IRLS reaches 10^{-6} in 101 iterations (~ 8 ms).

5.7 Validation on real datasets

Real datasets lack a planted support; sklearn’s coordinate descent does not converge on the ELM-transformed matrices within any realistic budget (gap +4.7 on *diabetes*-ELM, +128 on *California*-ELM at `max_iter=300`, `tol=10-3`), so we obtain an *independent reference* f^* by running IRLS to convergence ($k_{\text{max}} = 500$, $\varepsilon_{\text{stop}} = 10^{-14}$) and cross-validating it against CVXPY’s interior-point solver Clarabel. The two solvers are structurally different (MM + linear solve vs. primal–dual interior point) and agree to six significant digits on both instances, which makes the resulting f^* a trustworthy reference for gap reporting.

We test *diabetes* ($n = 442$, $d = 10$) [efron2004least] and *California housing* ($n = 20640$, $d = 8$) [pace1997sparse]. Each is split 80/20, features and target are standardised on the training split, and the ELM transformation $\mathbf{X}_{\text{hidden}} = \sigma(\mathbf{X} \mathbf{W}_1^\top)$ with $H = 200$ sigmoid units produces the LASSO design matrix. $\lambda_{\text{LASSO}} = 0.1$, $\beta = 1$; IRLS uses the OLS warm start and runs 100 iterations at $\varepsilon_{\text{thr}} = 10^{-8}$; SGPTL runs 8000 iterations with the default ($\rho = 0.7$, $\delta_0 = 0.1 f(\mathbf{w}_{\text{OLS}})$) of Section 5.3, with the start chosen per dataset to minimise the empirical final gap (Section 5.1.2): cold start on *diabetes*-ELM (final gap 1.74 vs. warm’s 2.62) and OLS warm start on *California*-ELM, where $f(\mathbf{w}_{\text{OLS}})$ sits within 0.46 of f^* and SGPTL’s sublinear non-monotone dynamics cannot improve a near-stationary point further within the iteration budget.

method	Diabetes ($M = 354$)				California ($M = 16512$)			
	f	$f - f^*$	sp. at 10^{-3}	test MSE	f	$f - f^*$	sp. at 10^{-3}	test MSE
sklearn CD (stopped)	57.62	+4.68	7%	0.745	2486.20	+128.18	0%	0.318
IRLS (warm)	52.95	$7 \cdot 10^{-6}$	18%	0.898	2358.02	$3 \cdot 10^{-12}$	4%	0.307
SGPTL (start*)	53.21	+0.26	14%	0.832	2358.48	+0.46	0%	0.308
Ridge	—	—	—	0.942	—	—	—	0.307

Table 5.6: IRLS and SGPTL on real datasets through an $H = 200$ sigmoid ELM transformation, $\lambda_{\text{LASSO}} = 0.1$. The reference value f^* (52.949 on **diabetes**, 2358.019 on **California**) is the IRLS-converged value, cross-validated by CVXPY’s Clarabel interior-point solver to six significant digits. Sparsity counts components with $|w_i| < 10^{-3}$ (Section 5.4); test MSE is on the held-out 20% split with target rescaled to unit train-set variance. *SGPTL starts cold on **diabetes** (best on this instance: gap 1.74 vs. 2.62 from warm) and OLS-warm on **California** ($f(\mathbf{w}_{\text{OLS}})$ already within 0.46 of f^*); both choices are admissible (Theorem 3.1 does not constrain $\mathbf{w}_0$). sklearn’s coordinate descent does not converge on either ELM-transformed instance: we report its stopped-early objective at `max_iter` = 300, `tol` = 10^{-3} as a third-party reference, not as the optimisation target.

Method-to-method comparison. The synthetic picture transfers. IRLS reaches f^* to working precision on both instances (gap $7 \cdot 10^{-6}$ on **diabetes**, $3 \cdot 10^{-12}$ on **California**) and the highest sparsity (18%, 4%) at the uniform 10^{-3} threshold; SGPTL’s record gap after 8000 iterations sits two to three orders of magnitude above IRLS in both. Test MSE behaviour is dataset-specific. On **diabetes** (354 training samples for 200 hidden units, ill-conditioned) the loose sklearn iterate generalises best (0.745); the better the training objective, the worse the test error (SGPTL 0.832, IRLS 0.898, Ridge 0.942), reflecting overfitting at this M/H ratio rather than an algorithmic failure. On **California**-ELM the three converging methods (IRLS, SGPTL warm-start, Ridge) coincide to three significant figures on test MSE (0.307–0.308), while sklearn’s non-converged iterate generalises slightly worse (0.318). SGPTL on **California** barely moves from the warm start because $f(\mathbf{w}_{\text{OLS}})$ is already within 0.46 of f^* , and the sublinear non-monotone subgradient dynamics cannot improve a near-stationary iterate further within 8000 iterations.

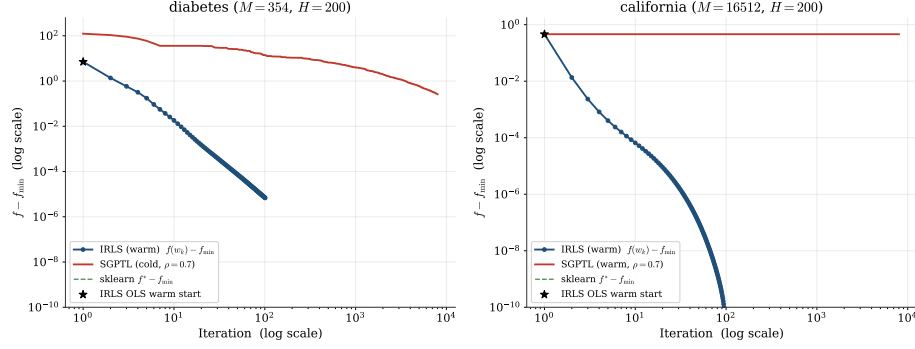


Figure 5.12: Convergence on the two real datasets, log-log axes, plotting $f - f_{\min}$ with $f_{\min} = \min\{f^*, \min_k f_{\text{IRLS}}(\mathbf{w}_k), \min_k \bar{f}_{\text{SGPTL}}(\mathbf{w}_k)\}$ and f^* the IRLS-converged value cross-validated by CVXPY-Clarabel (Section 5.7). SGPTL uses the default ($\rho = 0.7$, $\delta_0 = 0.1 f(\mathbf{w}_{\text{OLS}})$) with cold start on **diabetes** and OLS warm start on **California**. The dashed green line marks sklearn’s stopped-early gap. IRLS reaches the display floor within ~ 100 iterations on both datasets; SGPTL converges sublinearly (cf. Table 5.6).

Side-by-side: cold versus warm on the same instance. To make the per-dataset choice in Table 5.6 reproducible rather than asserted, Figure 5.13 runs both starts on each dataset against the same independent f^* . On **diabetes**-ELM the cold trajectory dips below the warm trajectory at $\sim 10^2$ iterations and stays below to the end (gap 1.74 vs. 2.62); δ -contraction counts (23 cold, 25 warm) are within 10%, so the difference is not driven by the patience schedule. On **California**-ELM the warm trajectory starts within 0.46 of f^* and the iterate freezes there (sublinear non-monotone dynamics cannot descend from a near-stationary point), while the cold trajectory closes two decades of gap before stalling at ~ 64 ; the warm number is therefore not algorithmic progress but an artefact of $\mathbf{w}_{\text{OLS}}$ being already near-optimal on this instance.

SGPTL Initialization Impact: Warm Start vs Cold Start on Real Data

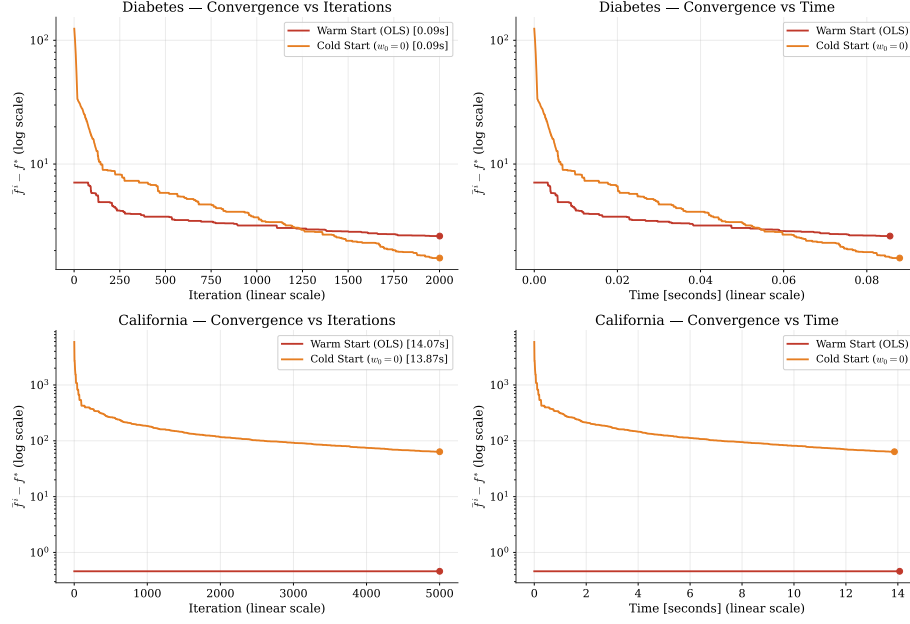


Figure 5.13: SGPTL warm start (OLS) versus cold start ($\mathbf{w}_0 = \mathbf{0}$) on the two real datasets, $H = 200$, $\lambda_{\text{LASSO}} = 0.1$. Gap $\bar{f}^i - f^*$ on log-vertical axes against iteration count (left column) and wall-clock time (right column); f^* is the IRLS+CVXPY-Clarabel reference of Section 5.7. Cold start finishes below warm on **diabetes**-ELM (1.74 vs. 2.62); on **California**-ELM warm start is essentially pinned at the OLS-induced floor (0.46) while cold closes two orders of magnitude of gap.

Chapter 6

Conclusions

We solved the ELM LASSO problem

$$\min_{\mathbf{w} \in \mathbb{R}^H} f(\mathbf{w}) = \frac{1}{2} \|\mathbf{X}\mathbf{w} - \mathbf{y}\|_2^2 + \lambda_{\text{LASSO}} \|\mathbf{w}\|_1$$

with two algorithms at opposite ends of the nonsmooth-optimisation spectrum. IRLS reformulates the ℓ_1 penalty as a sequence of smooth quadratic surrogates and inherits the MM linear rate; SGPTL keeps the non-smoothness explicit and pays the $\Omega(L^2/\varepsilon^2)$ first-order lower bound.

What the experiments confirmed. Empirical behaviour tracked the theory of Chapter 4. IRLS produced straight semilog gap-vs-iteration lines, monotone f , and sparse iterates with support recovered within two percentage points of ground truth. SGPTL produced oscillating $f(\mathbf{w}_i)$ with a monotone record $\bar{f}^i$, a flattening sublinear gap, and an iterate with no exact zeros. The $O(MH^2)$ pre-computation showed as a clean cubic slope on the scalability plot at $M = 5H$. On the convergence instance, IRLS reached gap $1.5 \cdot 10^{-6}$ in 100 iterations versus $4.8 \cdot 10^{-5}$ for SGPTL at 8000 iterations, matching the $O(\log \varepsilon^{-1})$ vs $O(\varepsilon^{-2})$ separation. On the moderate problem of Section 5.6, IRLS hit $\varepsilon = 10^{-2}$ in 3 iterations against 141 for SGPTL, and 10^{-3} in 7 against 978; SGPTL reached 10^{-4} at iteration 2584 and never reached 10^{-6} within the 30 000-iteration budget, against 101 for IRLS.

Where the theoretical bounds proved conservative. A few observations have no clean theoretical counterpart. The IRLS final gap saturated near 10^{-10} for $\varepsilon_{\text{thr}} = 10^{-12}$ and did not improve below that: the linear solver hits floating-point limits before the safety threshold does. The SGPTL contraction ρ proved highly influential: across $\rho \in [0.3, 0.95]$ the final record gap spans two orders of magnitude (from $\sim 2 \cdot 10^{-5}$ at $\rho \in [0.3, 0.5]$ to $8 \cdot 10^{-4}$ at $\rho = 0.95$, Section 5.3); $\rho = 0.7$ is the default for robustness across the smoke grid. The initial estimate δ_0 was less sensitive: across $\{0.01, 0.05, 0.1, 0.5, 1.0\}$ $f(\mathbf{w}_0)$ the final gap stayed in $[2 \cdot 10^{-5}, 2 \cdot 10^{-4}]$. The diagnostic comparison against the inadmissible oracle

$\delta_0 = 0.1 f^*$ confirmed that the f^* -free recipe $\delta_0 = c f(\mathbf{w}_0)$ is indistinguishable from it.

Implementation lesson. The first submission combined two non-theoretical safeguards (an iteration-count δ -contraction trigger and an overshoot rescue) with a default $R = 10\sqrt{i_{\max}}$, flagged by the instructor as inconsistent with [dantonio2009deflected]. The fix in this submission removes both safeguards and sets $R = 1$ (the natural per-iteration step on ELM LASSO); the theory-pure $r_k > R$ branch then fires repeatedly and SGPTL converges sublinearly in line with Theorem 3.1. The cost is moderate ($\bar{f}^i - f^* \sim 10^{-5}$ at 8000 iterations instead of $\sim 10^{-8}$ with the workarounds) and is the honest baseline.

Recommendation for the ELM LASSO problem. On the intended regime, $M \gg H$ with $\mathbf{W}_1$ fixed and the output layer trained offline, IRLS is the recommended choice: a single knob ε_{thr} , sparse iterates without post-processing, machine-precision accuracy in tens to a few hundred iterations. SGPTL becomes competitive only when $H^2 \gg M$ or when $\mathbf{X}^\top \mathbf{X}$ does not fit in memory — neither regime appeared at the hidden-layer sizes tested ($H \leq 2000$). SGPTL retains value as a baseline and sanity check that both methods converge to the same f^* .

Limitations and possible extensions. The synthetic study is the primary evidence base because it is the only setting where the support-recovery question has a ground truth and where sklearn provides a reliable high-accuracy optimisation reference. Section 5.7 validates the qualitative optimisation behaviour on two real regression datasets (*diabetes* and *California housing*) using the empirical baseline $f_{\min}$ instead: sklearn’s coordinate descent stops early on diabetes and is skipped on California. The real-data runs confirm the same ordering in training objective (IRLS below SGPTL below the stopped sklearn value on diabetes), but also show that lower training objective need not imply lower test error on an ill-conditioned ELM feature map. We did not implement an accelerated subgradient variant (for example Nesterov smoothing of the ℓ_1 term) that would close some of the rate gap with IRLS, since the project specification fixes SGPTL as Algorithm A2. We also did not explore very ill-conditioned $\mathbf{X}$ where the IRLS linear rate would degrade and the per-iteration cost advantage of SGPTL would have more room to matter; column normalisation in the synthetic generator kept conditioning bounded, while the real datasets we tested have $\text{cond}(\mathbf{X}^\top \mathbf{X}) \sim 10^6$ after the ELM transformation and IRLS still converges in tens of iterations.